CFA Level I Mock Exam 8.2 - Mark Scheme

Each question lists the correct choice and the original feedback for all three choices. Source excerpts accompany explanations where equations are PDF artwork.

Question 1

An analyst gathers the following information about three companies and their common shares:

	Company 1	Company 2	Company 3
Market price per share	$34	$12	$56
Book value of shareholders' equity (in millions)	$452	$302	$98
Number of shares outstanding (in millions)	45	105	12

Which company has the lowest price-to-book value per share?

Correct answer: A

A. Company 1 - Correct

Correct Answer Feedback: Correct because price to book value per share (P/B) = Market price per share / (Book value of shareholders' equity / shares outstanding). In Company 1's case, P/B = $34 / ($452 / 45) ≈ 34 / 10.044 ≈ 3.385.

B. Company 2 - Incorrect

Incorrect Answer Feedback: Incorrect because Company 2's P/B = $12 / ($302 / 105) ≈ 12 / 2.876 ≈ 4.172, which is higher than Company 1's P/B of 3.385. However, a candidate might select this if they incorrectly select Company 2 based on the fact it has the lowest book value / share (2.876).

C. Company 3 - Incorrect

Incorrect Answer Feedback: Incorrect because Company 3's P/B = $56 / ($98 / 12) ≈ 56 / 8.1667 ≈ 6.857, which is higher than Company 1's P/B of 3.385. However, a candidate might select this based on the fact that it has the lowest book value of shareholders' equity ($98 million).

Question 2

Which of the following is an example of an event-driven hedge fund strategy?

Correct answer: A

A. Buying the stock of a company that is spinning off a subsidiary - Correct

Correct Answer Feedback: Correct because it describes a special situations investment strategy, a type of event-driven investing. Special situations strategies focus on opportunities to get involved in the equity of companies that are engaged in restructuring activities other than mergers, acquisitions, or bankruptcy. These activities include security issuance or repurchase, special capital distributions, rescue finance, and asset sales/spin-offs.

B. Selling stock options to earn premiums and benefit from limited market movements - Incorrect

Incorrect Answer Feedback: Incorrect because it describes a relative value (short volatility) strategy. Relative value funds seek to profit from a pricing discrepancy, an unusual short-term relationship, between related securities. A short-volatility strategy (a type of relative value strategy) involves selling options to earn the premiums and benefit from calm markets but can experience significant losses during unexpected periods of market stress.

C. Taking long positions in undervalued stocks and short positions in overvalued stocks - Incorrect

Incorrect Answer Feedback: Incorrect because it describes an equity market neutral strategy. Market neutral strategies use quantitative (technical) and fundamental analysis to identify under- and overvalued equity securities. Hedge funds take long positions in securities they have identified as undervalued and short positions in securities they have identified as overvalued.

Question 3

Contracts entered into at one point in time that require both parties to engage in a transaction at a later point in time on terms agreed upon at the start are called:

Correct answer: C

A. options contracts. - Incorrect

Incorrect Answer Feedback: Incorrect because an option contract (which is also referred to as a contingent claim), is a derivative in which the outcome or payoff is dependent upon the outcome or payoff of an underlying asset. Although this characteristic is also associated with forward commitments, a contingent claim has come to be associated with a right, but not an obligation, to make a final payment contingent on the underlying. Given that the holder of the contingent claim has a choice, the term contingent claim has become synonymous with the term option.

B. contingent claims. - Incorrect

Incorrect Answer Feedback: Incorrect because a contingent claim is a derivative in which the outcome or payoff is dependent upon the outcome or payoff of an underlying asset. Although this characteristic is also associated with forward commitments, a contingent claim has come to be associated with a right, but not an obligation, to make a final payment contingent on the underlying.

C. forward commitments. - Correct

Correct Answer Feedback: Correct because forward commitments are contracts entered into at one point in time that require both parties to engage in a transaction at a later point in time (the expiration) on terms agreed upon at the start.

Question 4

A portfolio manager creates the following two-security portfolios:

Security	Security Weight	Expected Standard Deviation
A	30%	30%
B	70%	25%

If the correlation of returns between the two securities is 0.60, the expected standard deviation of the portfolio is closest to:

Correct answer: B

A. 16.8%. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses standard deviation instead of variance in the first part of the equation and did not take the square root = (0.3)2(0.30) + (0.7)2(0.25) + 2(0.3)(0.7)(0.60)(.30)(.25) = (0.027) + (0.1225) + (0.0189) = 0.16815 = 16.8%.

B. 24.0%. - Correct

Correct Answer Feedback: Correct because = = = = 0.240052078 = 24.0%.

Incorrect Answer Feedback: Incorrect because it uses standard deviation instead of variance in the first part of the equation and did not take the square root

$\sigma_{port} \neq w_2^1\sigma_1 + w_2^2\sigma_2 + 2w_1w_2\sigma_1\sigma_2\rho_{1,2}$ =

(0.3)^2(0.30) + (0.7)^2(0.25) + 2(0.3)(0.7)(0.60)(.30)(.25) =

(0.027) + (0.1225) + (0.0189) =

0.16815 = 16.8%.

Figure 1: Source equation for question 4, option A

Correct Answer Feedback: Correct because

$$\sigma_{port} = \sqrt{w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2w_1w_2\rho_{1,2}\sigma_1\sigma_2} =$$

$$\sigma_{port} = \sqrt{(0.3)^2(0.30)^2 + (0.7)^2(0.25)^2 + 2\left(0.3\right)\left(0.7\right)\left(0.6\right)\left(0.30\right)\left(0.25\right)}$$

=

$$\sigma_{port} = \sqrt{(0.09)(0.09) + (0.49)(0.0625) + 0.0189} =$$

$$\sigma_{port} = \sqrt{0.057625} = 0.240052078 = 24.0\%.$$

Figure 2: Source equation for question 4, option B

C. 26.5%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is simply the weighted average standard deviation. 0.30(0.3) + 0.70(0.25) = 0.265, or 26.5%.

Question 5

With respect to an IPS, which of the following best describes a relative risk objective for the client's portfolio?

Correct answer: A

A. An annual tracking risk of less than 2 percent - Correct

Correct Answer Feedback: Correct because for risk relative to a benchmark, the measure could be tracking risk, or tracking error. Further, an expected tracking risk of 2% would imply a return within 4% of the index return approximately 95% of the time. Remember that tracking risk is stated as a one standard deviation measure.

B. A 12-month 95% value at risk less than €1,000,000 - Incorrect

Incorrect Answer Feedback: Incorrect because measures of absolute risk include the variance or standard deviation of returns and value at risk. For example, the 12-month 95% value at risk of the portfolio must be no more than ¥1 billion.

C. Maintaining at least €10,000 in cash for planned monthly withdrawals - Incorrect

Incorrect Answer Feedback: Incorrect because maintaining cash is an IPS constraint, not a risk objective.

Question 6

A mutual fund that does not have a sales charge to purchase or redeem shares in the fund is best described as a(n):

Correct answer: C

A. exchange-traded fund. - Incorrect

Incorrect Answer Feedback: Incorrect because an exchange-traded fund is not classed as a mutual fund. Exchange-traded funds (ETFs) are investment funds that trade on exchanges (similar to individual stocks) and are generally structured as open-end funds. While ETFs are structured similarly to open-end mutual funds, some key differences exist between the two products.

B. open-end fund. - Incorrect

Incorrect Answer Feedback: Incorrect because the decision on a fund's load (the charge for purchasing or redeeming shares) is separate from the decision on whether it is an open-end or closed-end fund. The fund can be set up as an open-end fund or a closed-end fund. If it is an open- end fund, it will accept new investment money and issue additional shares at a value equal to the net asset value of the fund at the time of investment. An open-end fund can be either a load or a no-load fund.

C. no-load fund. - Correct

Correct Answer Feedback: Correct because, in addition to open-end or closed-end funds, mutual funds can be classified as load or no-load funds. The primary difference between the two is whether the investor pays a sales charge (a 'load') to purchase, hold, or redeem shares in the fund. In the case of the no-load fund, there is no fee for investing in the fund or for redemption but there is an annual fee based on a percentage of the fund's net asset value.

Question 7

Using the following information and assuming coupons are paid annually, the G-spread of the Steel Co. bond is closest to:

Bond	Maturity	Coupon	Price
Steel Co.	2 Years	5.00%	101.70
Treasury bond	2 Years	4.00%	100.50

Correct answer: A

A. 36 bps. - Correct

Correct Answer Feedback: Correct because the yield for the Steel Co. bond is calculated as The yield for the Treasury bond is calculated as G-spread is calculated as the yield difference between the Steel Co. Bond and the Treasury bond: 4.0974% – 3.7359% = 36.15%, or 36 bps.

B. 94 bps. - Incorrect

Incorrect Answer Feedback: Incorrect because it is calculated as current yield difference: (5/101.70) – (4/100.50) = 4.9164% – 3.9801% = 0.9363%, i.e., 94 bps.

C. 100 bps. - Incorrect

Incorrect Answer Feedback: Incorrect because it is calculated as coupon rate difference: 5% – 4% = 1%.

Question 8

Which of the following statements regarding forward contracts is most accurate?

Correct answer: B

Correct Answer Feedback: Correct because the yield for the Steel Co. bond is calculated as

$$\frac{5}{(1+r)} + \frac{5+100}{(1+r)^2} = 101.70$$

$$r = 4.0974\%$$

The yield for the Treasury bond is calculated as

$$\frac{4}{(1+r)} + \frac{4+100}{(1+r)^2} = 100.50$$

$$r = 3.7359\%$$

G-spread is calculated as the yield difference between the Steel Co. Bond and the Treasury bond: 4.0974% – 3.7359% = 36.15%, or 36 bps.

Figure 3: Source equation for question 7, option A

A. The value of a short forward contract is the spot price minus the forward price at expiration - Incorrect

Incorrect Answer Feedback: Incorrect because, at expiration, the value to the short party is the forward price minus the spot price at expiration.

B. The value of a long forward contract increases as the spot price of the underlying increases - Correct

Correct Answer Feedback: Correct because the value of a forward contract is the spot price of the underlying asset minus the present value of the forward price. As the spot price of the underlying increases, so does the value of the long forward contract.

C. The forward price is the spot price compounded at the risk premium over the life of the contract - Incorrect

Incorrect Answer Feedback: Incorrect because the forward price is the spot price compounded at the risk-free rate over the life of the contract, not the risk premium.

Question 9

Which of the following features of an established hedge fund is most likely designed to allow orderly liquidation of positions to meet redemptions?

Correct answer: A

A. Notice period - Correct

Correct Answer Feedback: Correct because notice periods provide an opportunity for the hedge fund manager to liquidate a position in an orderly fashion without magnifying the losses.

B. Lockup period - Incorrect

Incorrect Answer Feedback: Incorrect because lockup periods, time periods when investors cannot withdraw their capital, provide the hedge fund manager the required time to implement and potentially realize a strategy's expected results. The hedge fund is already experiencing redemptions and thus, although a lockup period may help at the margin by delaying some redemptions that would otherwise need to be funded, a notice period for redemptions is more likely to assist the hedge fund manager to liquidate positions in an orderly manner.

C. Redemption fee - Incorrect

Incorrect Answer Feedback: Incorrect because funds sometimes charge redemption fees (typically payable to the remaining investors) to discourage redemption and to offset the transaction costs for remaining investors. The hedge fund is already experiencing redemptions and thus, although a redemption fee may help at the margin

by offsetting costs, a notice period for redemptions is more likely to assist the hedge fund manager to liquidate positions in an orderly manner.

Question 10

The type of index weighting system in which a stock split on one constituent security changes the weights on all the securities in the index is:

Correct answer: A

A. price weighting. - Correct

Correct Answer Feedback: Correct because a property unique to price-weighted indexes is that a stock split on one constituent security changes the weights on all the securities in the index.

B. equal weighting. - Incorrect

Incorrect Answer Feedback: Incorrect because this is a characteristic associated with a price- weighted index. An equal weighted index is rebalanced when its constituent securities' prices change in order to maintain equal-weighting. After the index is constructed and the prices of constituent securities change, the index is no longer equally weighted. Therefore, maintaining equal weights requires frequent adjustments (rebalancing) to the index.

C. value weighting. - Incorrect

Incorrect Answer Feedback: Incorrect because this is a characteristic associated with a price-weighted index. A stock split would not change the relative weight of a security in a value-weighted index since the total market value of the security would be unchanged after the split. In market-capitalization weighting, or value weighting, the weight on each constituent security is determined by dividing its market capitalization by the total market capitalization (the sum of the market capitalization) of all the securities in the index. Market capitalization or value is calculated by multiplying the number of shares outstanding by the market price per share.

Question 11

For European options, all else being equal, a higher exercise price results in a higher value for:

Correct answer: A

A. puts only. - Correct

Correct Answer Feedback: Correct because the value of a European call option is inversely related to the exercise price whereas the value of a European put option is directly related to the exercise price. Therefore, all else being equal, the higher the exercise price, the higher is the value of European puts and the lower is the value of European calls.

B. calls only. - Incorrect

Incorrect Answer Feedback: Incorrect because higher exercise price results in a higher value for puts only (not calls), as explained in the rationale for the correct answer.

C. both puts and calls. - Incorrect

Incorrect Answer Feedback: Incorrect because higher exercise price results in a higher value for puts only (not calls), as explained in the rationale for the correct answer.

Question 12

All else being equal, which of the following factors is most likely associated with a narrowing corporate bond yield spread?

Correct answer: C

A. An economic slowdown - Incorrect

Incorrect Answer Feedback: Incorrect because weakening economic conditions will push investors to desire a greater risk premium and drive overall credit spreads wider.

B. A negative earnings release by the issuer - Incorrect

Incorrect Answer Feedback: Incorrect because a negative earnings release by the issuer is associated with a widening of corporate bond yield spreads. Good news increases the attractiveness of buying and holding bonds issued by the corporation, which raises bond prices and narrows spreads; bad news will have the opposite effect.

C. Greater market demand for bonds relative to supply - Correct

Correct Answer Feedback: Correct because greater demand for bonds relative to supply are associated with narrower yields. In periods of heavy new issue supply, credit spreads will widen if there is insufficient demand. In periods of high demand for bonds, spreads will move tighter.

Question 13

Which of the following is least likely one of the main benefits of derivative markets? Derivative markets:

Correct answer: B

A. reveal prices and volatility of the underlying assets. - Incorrect

Incorrect Answer Feedback: Incorrect. One of the main purposes of derivative markets is price discovery.

B. exhibit lower volatility compared with the spot market. - Correct

Correct Answer Feedback: Correct. Derivative markets are not necessarily more or less volatile than spot markets. And excessive risk taking and use of leverage in derivative markets may contribute to market stress.

C. enable companies to more easily practice risk management. - Incorrect

Incorrect Answer Feedback: Incorrect. One of the main purposes of derivative markets is risk management

Question 14

Which of the following statements regarding indexes for alternative investments is most accurate?

Correct answer: A

A. Hedge funds determine whether their results will be included in an index. - Correct

Correct Answer Feedback: Correct. Hedge fund companies are not required to report their performance to any party other than their investors. As a result, rather than index providers determining the constituents, the constituents determine the index.

B. Commodity index returns are based on changes in the prices of the underlying commodities - Incorrect

Incorrect Answer Feedback: Incorrect. Commodity indexes are based upon futures contracts, not prices of the underlying commodities themselves.

C. Commodity indexes consist of a market-weighted basket of one or more specific commodities - Incorrect

Incorrect Answer Feedback: Incorrect. Commodity indexes consist of futures contracts. There is no market weighting mechanism.

Question 15

Privately held convertible debt is most likely used by companies in which of the following life cycle stages?

Correct answer: A

A. Seed - Correct

Correct Answer Feedback: Correct because privately held convertible debt is relied on in the seed or start-up stages of the corporate life cycle.

B. Growth - Incorrect

Incorrect Answer Feedback: Incorrect because privately held convertible debt is relied on in the seed or start-up stages of the corporate life cycle, not in the growth stage.

C. Maturity - Incorrect

Incorrect Answer Feedback: Incorrect because privately held convertible debt is relied on in the seed or start-up stages of the corporate life cycle, not in the maturity stage.

Question 16

With respect to risk-adjusted return measures, the information ratio is calculated as Jensen's alpha divided by a security's:

Correct answer: C

A. total variance. - Incorrect

Incorrect Answer Feedback: Incorrect because the information ratio for a security is Jensen's alpha divided by nonsystematic variance, not total variance.

B. systematic variance. - Incorrect

Incorrect Answer Feedback: Incorrect because the information ratio for a security is Jensen's alpha divided by nonsystematic variance, not systematic variance.

C. nonsystematic variance. - Correct

Correct Answer Feedback: Correct because the information ratio for a security is Jensen's alpha divided by nonsystematic variance. The information ratio, (i.e., alpha divided by nonsystematic risk), measures the abnormal return per unit of risk added by the security to a well-diversified portfolio. The larger the information ratio is, the more valuable the security.

Question 17

Which of the following derivative contracts has a hard commodity underlying?

Correct answer: C

A. Cattle futures - Incorrect

Incorrect Answer Feedback: Incorrect because cattle futures are an example of a derivative contract with a soft commodity underlying. Soft commodities are agricultural products, such as cattle and corn.

B. Soybean futures - Incorrect

Incorrect Answer Feedback: Incorrect because soybean options (and futures) are an example of a derivative contract with an agricultural, or soft, commodity underlying.

C. Aluminum futures - Correct

Correct Answer Feedback: Correct because aluminum futures are an example of a metals contract, which is a derivative with a hard commodity underlying.

Question 18

Which of the following is an exclusive feature of private equity funds as opposed to hedge funds?

Correct answer: C

A. Hurdle rates - Incorrect

Incorrect Answer Feedback: Incorrect because the partnership agreement usually specifies that the performance fee is earned only after the fund achieves a return known as a hurdle rate. The hurdle rate is a minimum rate of return. Hurdle rates are less common for hedge funds but do appear from time to time.

B. Performance fees - Incorrect

Incorrect Answer Feedback: Incorrect because on top of the management fee, a performance fee (also referred to as an incentive fee or carried interest) is applied based on excess returns. The partnership agreement usually specifies that the performance fee is earned only after the fund achieves a return known as a hurdle rate. The hurdle rate is a minimum rate of return. Hurdle rates are less common for hedge funds but do appear from time to time. As for hedge fund managers, they generally accrue an incentive fee on their quarterly performance, but they typically crystalize (realize) their incentive fee once annually.

C. Management fees based on committed capital - Correct

Correct Answer Feedback: Correct because the management fee is typically based on committed capital, not invested capital; the committed-capital basis for management fees is an important distinction from hedge funds, whose management fees are based on assets under management (AUM).

Question 19

An investor evaluating a company's common stock has gathered the following information:

Current dividend per share	$2.40
Dividend growth rate expected during Years 1 to 2	20%
Dividend growth rate expected from Year 3 onward	4%
Company's WACC	13%
Required rate of return on equity	15%

The intrinsic value per share of this common stock is closest to:

Correct answer: C

A. $24.86. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses incorrect computations for D , 1 D , and D by making mistakes in accounting for time periods. 2 3 V = = $2.09 + $2.18 + $20.59 = $24.86

Incorrect Answer Feedback: Incorrect because it uses incorrect computations for D_1, D_2, and D_3 by making mistakes in accounting for time periods.

$$V = \frac{2.40}{1.15} + \frac{2.40(1.20)}{1.15^2} + \frac{2.40(1.20)(1.04)}{(0.15-0.04)} \times \frac{1}{1.15^2}$$

$$= \$2.09 + \$2.18 + \$20.59 = \$24.86$$

Figure 4: Source equation for question 19, option A

B. $26.59. - Incorrect

Incorrect Answer Feedback: Incorrect because the terminal value at the end of Year 2 is discounted with n = 3 instead of n = 2. V = = $2.50 + $2.61 + $21.48 = $26.59

Incorrect Answer Feedback: Incorrect because the terminal value at the end of Year 2 is discounted with $n = 3$ instead of $n = 2$.

$$V = \frac{2.40(1.20)}{1.15} + \frac{2.40(1.20)^2}{1.15^2} + \frac{2.40(1.20)^2(1.04)}{(0.15-0.04)} \times \frac{1}{1.15^3}$$

$$= \$2.50 + \$2.61 + \$21.48 = \$26.59$$

Figure 5: Source equation for question 19, option B

C. $29.82. - Correct

Correct Answer Feedback: Correct because using the two-stage dividend discount model: V = 0 V = intrinsic value of a stock 0 D = expected dividend in year t t r = required rate of return on the stock D = Current dividend per share = $2.40 t V = 0 = $2.50 + $2.61 + $24.71 = $29.82

Correct Answer Feedback: Correct because using the two-stage dividend discount model:

$$V_0 = \frac{D_1}{(1+r)^1} + \cdots + \frac{D_n}{(1+r)^n} + \frac{P_n}{(1+r)^n}$$

V_0 = intrinsic value of a stock

D_t = expected dividend in year t

r = required rate of return on the stock

D_t = Current dividend per share = $2.40

$$V_0 = \frac{2.40(1.20)}{1.15} + \frac{2.40(1.20)^2}{1.15^2} + \frac{2.40(1.20)^2(1.04)}{(0.15-0.04)} \times \frac{1}{1.15^2}$$

$$= \$2.50 + \$2.61 + \$24.71 = \$29.82$$

Figure 6: Source equation for question 19, option C

Question 20

For parallel shifts in the benchmark yield curve, key rate durations indicate the same interest rate sensitivity as:

Correct answer: A

A. effective duration. - Correct

Correct Answer Feedback: Correct because for parallel shifts in the benchmark yield curve, yields at all maturity segments of the curve move the same in both direction and magnitude and the curve maintains its shape (no flattening or steepening). As a result, key rate durations at all maturity segments will indicate the same interest

rate sensitivity as effective duration, the measure of interest rate risk in terms of a parallel shift in the benchmark yield curve.

B. modified duration. - Incorrect

Incorrect Answer Feedback: Incorrect because modified duration is a yield duration (rather than a curve duration) statistic that does not measure a bond's sensitivity to a change in the benchmark yield curve.

C. Macaulay duration. - Incorrect

Incorrect Answer Feedback: Incorrect because Macaulay duration is a yield duration (rather than a curve duration) statistic that does not measure a bond's sensitivity to a change in the benchmark yield curve.

Question 21

Adding alternative investments to a portfolio of traditional investments most likely increases portfolio:

Correct answer: C

A. liquidity. - Incorrect

Incorrect Answer Feedback: Incorrect because investors are often attracted to alternative investments when seeking greater diversification and/or higher expected returns in exchange for what are often longer-term, illiquid investments in less efficient markets.

B. transparency. - Incorrect

Incorrect Answer Feedback: Incorrect because fund investment structures for alternative investments differ substantially from traditional public equity and fixed-income fund or ETF investments. For example, alternative funds usually involve less frequent transparency on periodic returns and fund positions versus equity or fixed-income funds.

C. diversification. - Correct

Correct Answer Feedback: Correct because investors often consider alternative investments in pursuit of greater portfolio diversification.

Question 22

The binomial option pricing model always:

Correct answer: A

A. requires knowledge of the magnitude of possible up and down moves. - Correct

Correct Answer Feedback: Correct because the binomial model requires knowledge of what the upside case is (represented by S +) and what the downside case is (as represented by S –). 1 1 We cannot arbitrarily set these values at just anything. We will be required to know the values of S + and S –. 1 1

B. requires knowledge of the actual probabilities of possible up and down moves. - Incorrect

Incorrect Answer Feedback: Incorrect because the actual probabilities of the up and down moves do not matter.

C. enables the construction of a risk-free hedge consisting of one unit of each of the option and the underlying. - Incorrect

Incorrect Answer Feedback: Incorrect because, while the binomial model does enable the construction of a risk-free hedge consisting of the option and the underlying, the appropriate ratio needs to be calculated. One needs to derive the number of units of the underlying that will perfectly hedge one unit of the short call, and one unit of each is not the appropriate balance.

Question 23

A mortgage pass-through security most likely has:

Correct answer: B

A. stable cash flows. - Incorrect

Incorrect Answer Feedback: Incorrect because the investors in these securities receive the principal and interest payments from the assets (often loans) in the pool as they are “passed through” the legal entity. Pass-through securities pass through the payments proportionally across the different tranches and redistribute the payment risks to investors seeking differing levels of risk exposures. The payments investors receive vary and depend directly on the overall credit risk of the pool and payment patterns.

B. an uncertain average life. - Correct

Correct Answer Feedback: Correct because prepayments for fixed-rate mortgages will vary depending on interest rates. When interest rates decline, actual prepayments will be higher than forecasted because homeowners will refinance at lower interest rates. Mortgage refinancing is particularly common in the US given the absence of loan prepayment penalties, which are more prevalent in other countries. Thus, the maturity of an MBS will be shorter than what was anticipated at the time of purchase. Contraction risk is the risk that the borrower repays the principal or a portion of the principal in a shorter period of time than the contractually agreed scheduled payment, reducing the amount of future payments the investor receives. For investors, this has two adverse consequences. First, investors must reinvest the proceeds at lower interest rates. Second, the prepayment option reduces the potential price appreciation for the bond. When interest rates increase, actual prepayments will be lower than forecasted because homeowners will become less likely to refinance their mortgages and may even delay new home purchases. Thus, the maturity of an MBS will be longer than what was anticipated at the time of purchase. Extension risk is the risk that the borrower repays the principal or a proportion of the principal in a longer time period than the contractually agreed scheduled payment. For investors, this has adverse consequences: Higher interest rates reduce the value of the cash flows investors receive. Therefore, payments the investors receive will be discounted at a higher interest rate, and the extension stretches out the payments the investors receive.

C. greater cash flows than the underlying pool of mortgages. - Incorrect

Incorrect Answer Feedback: Incorrect because the monthly cash flows of a mortgage pass- through security are less than the monthly cash flow of the underlying pool of mortgages by an amount equal to the servicing and other administrative fees. The cash flows of a mortgage pass- through security depend on the monthly cash flows of the underlying pool of mortgages. The amount and the timing of the cash flows from the pool of mortgages cover both the cash flow passed through to the security holders and administrative charges for servicing the pool. These administrative charges include collecting monthly payments from borrowers, forwarding proceeds to owners of the loan, sending payment notices to borrowers, maintaining records of the outstanding mortgage balance, initiating foreclosure proceedings if necessary, and providing tax information to borrowers when applicable. Additionally, the issuer or financial guarantor of the mortgage pass-through security may charge fees for guaranteeing the issue. All these fees are typically calculated as a portion of the mortgage rate.

Question 24

The price value of a bp for a 30-year, 1% annual coupon payment bond with a par value of €1,000 and a yield to maturity of 0.8% is closest to:

Correct answer: B

A. €2.58. - Incorrect

Incorrect Answer Feedback: Incorrect because the 1 bp reduction and increase is wrongly applied to the coupon rate, not the yield-to-maturity: PVBP = (1,002.58 – 997.42) / 2 = 2.58, whereby PV : FV = 1,000; PMT = 10; N = 30; I/Y = 0.99%; CPT PV = 1,002.58 and – PV : FV = 1,000; PMT = 10; N = 30; I/Y = 1.01%; CPT PV = 997.42. +

B. €2.74. - Correct

Correct Answer Feedback: Correct because: PVBP = [(full price calculated by lowering the yield-to-maturity by 1 bp) – (full price calculated by raising the yield-to-maturity by 1 bp)] / 2 PVBP = (1,055.90 – 1,050.42) / 2 = 2.74,

whereby PV : FV = 1,000; PMT = 10; N = 30; I/Y = 0.79%; CPT PV = 1,055.90 and – PV : FV = 1,000; PMT = 10; N = 30; I/Y = 0.81%; CPT PV = 1,050.42. +

C. €3.00. - Incorrect

Incorrect Answer Feedback: Incorrect because the 1 bp is wrongly multiplied with the 30-year tenor of the bond and the par value: PVBP = 30 × 0.01% × 1,000 = 3.00.

Question 25

Based on put-call parity, a fiduciary call can be created by combining:

Correct answer: A

A. a long call and a long risk-free zero-coupon bond. - Correct

Correct Answer Feedback: Correct because a fiduciary call is equal to a long call and a long risk-free zero-coupon bond.

B. a short put and a long risk-free zero-coupon bond. - Incorrect

Incorrect Answer Feedback: Incorrect because a fiduciary call is not equal to a short put and a long risk-free zero-coupon bond.

C. a long call and a short risk-free zero-coupon bond. - Incorrect

Incorrect Answer Feedback: Incorrect because a fiduciary call is not equal to a long call and a short risk-free zero-coupon bond.

Question 26

The Gordon growth model assumes that the dividend growth rate is:

Correct answer: A

A. less than the required rate of return. - Correct

Correct Answer Feedback: Correct because one of the assumptions of the Gordon model is: The dividend growth rate is strictly less than the required rate of return.

B. equal to the required rate of return. - Incorrect

Incorrect Answer Feedback: Incorrect because one of the assumptions of the Gordon model is: The dividend growth rate is strictly less than the required rate of return.

C. greater than the required rate of return. - Incorrect

Incorrect Answer Feedback: Incorrect because one of the assumptions of the Gordon model is: The dividend growth rate is strictly less than the required rate of return.

Question 27

When calculating portfolio duration as the weighted average duration of individual bonds, an analyst implicitly assumes that the yield curve:

Correct answer: B

A. flattens. - Incorrect

Incorrect Answer Feedback: Incorrect because this measure of portfolio duration implicitly assumes a parallel shift in the yield curve. In reality, interest rate changes frequently result in a steeper or flatter yield curve. Therefore, portfolio duration is the best measure of interest rate risk of a bond portfolio when there is a parallel shift in the yield curve, and not when the yield curve flattens.

B. experiences a parallel shift. - Correct

Correct Answer Feedback: Correct because the Macaulay and modified durations for the portfolio are calculated as the weighted average of the statistics for the individual bonds. The shares of overall portfolio market value are the weights. This measure of portfolio duration implicitly assumes a parallel shift in the yield curve. Therefore, portfolio duration is the best measure of the interest rate risk of a bond portfolio when there is a parallel shift in the yield curve.

C. steepens. - Incorrect

Incorrect Answer Feedback: Incorrect because portfolio duration implicitly assumes a parallel shift in the yield curve. In reality, interest rate changes frequently result in a steeper or flatter yield curve. Therefore, portfolio duration is the best measure of interest rate risk of a bond portfolio when there is a parallel shift in the yield curve, and not when the yield curve steepens.

Question 28

If interest rates are increasing, the effective duration of a bond with an embedded put option is:

Correct answer: A

A. less than the effective duration of an option-free bond. - Correct

Correct Answer Feedback: Correct because an embedded put option reduces the effective duration of the bond, especially when rates are rising.

B. the same as the effective duration of an option-free bond. - Incorrect

Incorrect Answer Feedback: Incorrect because an embedded put option reduces the effective duration of the bond, especially when rates are rising.

C. greater than the effective duration of an option-free bond. - Incorrect

Incorrect Answer Feedback: Incorrect because an embedded put option reduces the effective duration of the bond, especially when rates are rising.

Question 29

Which of the following embedded options most likely provides a right to the issuer of a bond?

Correct answer: B

A. Put feature - Incorrect

Incorrect Answer Feedback: Incorrect because the right to put the issue is an option granted to the bondholder to sell the bond back to the issuer at a predetermined price on specified dates before maturity.

B. Call feature - Correct

Correct Answer Feedback: Correct because the right to call the issue is beneficial to the issuer when interest rates fall.

C. Conversion provision - Incorrect

Incorrect Answer Feedback: Incorrect because the conversion provision grants the bondholder the right to convert the bond for a specified number of shares of common stock. Such a feature allows the bondholder to take advantage of favorable movements in the price of the issuer’s common stock.

Question 30

According to portfolio theory, combinations of the risk-free asset and a risky asset result in:

Correct answer: A

A. a capital allocation line. - Correct

Correct Answer Feedback: Correct because the combination of the risk-free asset and a risky asset can result in a better risk–return trade-off than an investment in only one type of asset because the risk-free asset has zero correlation with the risky asset. The combination is called the capital allocation line.

B. the Markowitz efficient frontier. - Incorrect

Incorrect Answer Feedback: Incorrect because the curve that lies above and to the right of the global minimum-variance portfolio is referred to as the Markowitz efficient frontier because it contains all portfolios of risky assets that rational, risk-averse investors will choose. That is, the Markowitz efficient frontier only contains risky assets, while the capital allocation line is a combination of the risk-free asset and a risky asset.

C. an investor's indifference curve. - Incorrect

Incorrect Answer Feedback: Incorrect because the capital allocation line (CAL), not the investor's indifference curve, is the combination of the risk-free asset and a risky asset. An indifference curve plots the combinations of risk–return pairs that an investor would accept to maintain a given level of utility (i.e., the investor is indifferent about the combinations on any one curve because they would provide the same level of overall utility). Indifference curves are thus defined in terms of a trade-off between expected rate of return and variance of the rate of return. Because an infinite number of combinations of risk and return can generate the same utility for the same investor, indifference curves are continuous at all points.

Question 31

If the annual market discount rate is 6%, the value of a 3-year bond that has a 7% coupon rate, has a maturity (par) value of $1,000, and pays interest annually is closest to:

Correct answer: B

A. $973.76. - Incorrect

Incorrect Answer Feedback: Incorrect because the answer is $1,026.73. ($973.76 is calculated by reversing the coupon rate and discount rate.)

B. $1,026.73. - Correct

Correct Answer Feedback: Correct because t PV = PMT / (1+r) , where t PV is the present value PMT is the payment = par value × coupon rate = $70 r is the market discount rate. Expanding this equation over three years results in a present value equal to:

Correct Answer Feedback: Correct because

$PV = PMT_t / (1+r)^t$, where

PV is the present value

PMT is the payment = par value × coupon rate = $70

r is the market discount rate.

Expanding this equation over three years results in a present value equal to:

$$\frac{\$70}{1.06^1} + \frac{\$70}{1.06^2} + \frac{\$1{,}070}{1.06^3} = \$66.04 + \$62.30 + \$898.39 = \$1,026.73$$

Figure 7: Source equation for question 31, option B

C. $1,049.17. - Incorrect

Incorrect Answer Feedback: Incorrect because the answer is $1,026.73. ($1,049.17 is derived by doubling the number of payments to 6, as if the bond were making semi- annual payments rather than annual.)

Question 32

Which of the following is an underlying assumption of yield to maturity as a measure of expected return for a bond?

Correct answer: C

A. The credit quality of the issuer remains the same - Incorrect

Incorrect Answer Feedback: Incorrect because although one of the assumptions of yield-to- maturity as a return measure is that there is no default by the issuer, it only assumes that the issuer does not default, meaning all interest and principal payments are made as scheduled. Although a change to the credit rating of the issuer may change the market price of the bond before maturity, the return measure is yield to maturity, meaning principal repayment is par at maturity, and credit ratings in the interim or at maturity will not impact that amount.

B. The bond is sold at the same price as the purchase price - Incorrect

Incorrect Answer Feedback: Incorrect because the yield-to-maturity at the time of purchase measures the investor's rate of return under three assumptions: (1) The investor holds the bond to maturity, (2) there is no default by the issuer, and (3) the coupon interest payments are reinvested at that same rate of interest. The assumptions require the bond to be held to maturity, not to be sold at the purchase price.

C. Coupons are reinvested at the same rate as the yield to maturity at purchase - Correct

Correct Answer Feedback: Correct because the yield-to-maturity at the time of purchase measures the investor's rate of return under three assumptions: (1) The investor holds the bond to maturity, (2) there is no default by the issuer, and (3) the coupon interest payments are reinvested at that same rate of interest.

Question 33

Which of the following measures is an example of a top-down driver for forecasting a firm's revenue?

Correct answer: B

A. Same-store sales growth - Incorrect

Incorrect Answer Feedback: Incorrect because same-store sales growth is an example of a capacity-based measure which is a bottom-up forecast object. Examples of bottom-up drivers for revenue forecasts include capacity-based measures. Forecasts, for example, in retailing, based on the number of stores and sales per store, or same-store sales growth (for stores that have been open for at least 12 months) and sales related to new-store openings.

B. Growth relative to GDP growth - Correct

Correct Answer Feedback: Correct because growth relative to GDP growth is an example of a top-down forecast object. Common top-down forecast objects include 'growth relative to GDP growth' and 'market growth and market share.'

C. Volumes and average selling prices - Incorrect

Incorrect Answer Feedback: Incorrect because examples of bottom-up drivers for revenue forecasts include the following: Volumes and average selling prices.

Question 34

Which of the following attributes is least likely to be a requirement for the existence of riskless arbitrage? The underlying security:

Correct answer: C

A. is relatively liquid. - Incorrect

Incorrect Answer Feedback: Incorrect because for riskless arbitrage to exist, the underlying security must be relatively liquid so it is easy to buy and sell at a low cost.

B. can be sold short. - Incorrect

Incorrect Answer Feedback: Incorrect because for riskless arbitrage to exist, the underlying security must be able to be short sold.

C. is a financial asset. - Correct

Correct Answer Feedback: Correct because riskless arbitrage can exist when the underlying security that can be arbitraged is either a financial or a non-financial security.

Question 35

Returns adjusted for the total risk of a portfolio relative to a benchmark are most appropriately measured using which of the following? 2

Correct answer: A

A. M - Correct

Correct Answer Feedback: Correct because M2 provides a measure of portfolio return that is adjusted for the total risk of the portfolio relative to that of some benchmark. Total risk is relevant for an investor when he or she holds a portfolio that is not fully diversified, which is not a desirable portfolio. In such cases, the Sharpe ratio and M2 are appropriate performance measures.

B. Treynor ratio - Incorrect

Incorrect Answer Feedback: Incorrect because the Treynor ratio is a simple extension of the Sharpe ratio and resolves the Sharpe ratio's first limitation by substituting beta (systematic risk) for total risk.

C. Jensen's alpha - Incorrect

Incorrect Answer Feedback: Incorrect because Jensen's alpha is based on systematic risk. Performance measures relative to beta risk—Treynor ratio and Jensen's alpha—are relevant when the investor holds a well-diversified portfolio with negligible diversifiable risk.

Question 36

A bond is currently selling for 102.31. A valuation model estimates the price will fall to 101.12 if interest rates increase by 20 bps and rise to 103.74 if interest rates decrease by 20 bps. Using these estimates, the effective duration of the bond is closest to:

Correct answer: B

A. 6.31. - Incorrect

Incorrect Answer Feedback: Incorrect because it has 103.74 in the denominator instead of 102.31:

Incorrect Answer Feedback: Incorrect because it has 103.74 in the denominator instead of 102.31:

$$\frac{103.74-101.12}{2\times103.74\times0.002} = 6.31$$

Figure 8: Source equation for question 36, option A

B. 6.40. - Correct

Correct Answer Feedback: Correct because the effective duration of a bond is EffDur = where PV , PV , and PV are the values of the bond – 0 + when the yield falls, under the current yield, and when the yield rises, respectively, and ΔCurve is the change in the benchmark yield curve. EffDur =

Correct Answer Feedback: Correct because the effective duration of a bond is

$$\text{EffDur} = \frac{(PV_-)-(PV_+)}{2\times(\Delta\text{Curve})\times(PV_0)}$$ where PV_-, PV_0, and PV_+ are the values of the bond when the yield falls, under the current yield, and when the yield rises, respectively, and ΔCurve is the change in the benchmark yield curve.

$$\text{EffDur} = \frac{103.74-101.12}{2\times102.31\times0.002} = 6.40$$

Figure 9: Source equation for question 36, option B

C. 6.48. - Incorrect

Incorrect Answer Feedback: Incorrect because it has 101.12 in the denominator instead of 102.31:

Incorrect Answer Feedback: Incorrect because it has 101.12 in the denominator instead of 102.31:

$$\frac{103.74-101.12}{2\times101.12\times0.002} = 6.48$$

Figure 10: Source equation for question 36, option C

Question 37

Which of the following cognitive errors is best classified as a processing error?

Correct answer: C

A. Confirmation bias - Incorrect

Incorrect Answer Feedback: Incorrect because it is a belief perseverance bias, and not a processing error. The belief perseverance biases discussed are conservatism, confirmation, representativeness, illusion of control, and hindsight.

B. Illusion of control bias - Incorrect

Incorrect Answer Feedback: Incorrect because it is a belief perseverance bias, and not a processing error. The belief perseverance biases discussed are conservatism, confirmation, representativeness, illusion of control, and hindsight.

C. Anchoring and adjustment bias - Correct

Correct Answer Feedback: Correct because the processing errors discussed are anchoring and adjustment, mental accounting, framing, and availability. Anchoring and adjustment bias refers to relying on an initial piece of information to make subsequent estimates, judgments, and decisions.

Question 38

A bond issued by the International Monetary Fund is best classified as a:

Correct answer: B

A. sovereign bond. - Incorrect

Incorrect Answer Feedback: Incorrect because sovereign bonds are issued by national governments. National governments issue bonds primarily for fiscal reasons—to fund spending when tax revenues are insufficient to cover expenditures. Bonds issued by national governments, are often referred to as sovereign bond or sovereigns.

B. supranational bond. - Correct

Correct Answer Feedback: Correct because supranational bonds are a form of often highly rated bonds issued by supranational agencies, also referred to as multilateral agencies. The most well-known supranational agencies are the International Bank for Reconstruction and Development (the World Bank), and the International Monetary Fund (IMF).

C. quasi-government bond. - Incorrect

Incorrect Answer Feedback: Incorrect because quasi-government bonds are issued by entities established by national governments to perform various functions. These organizations often have both public and private sector characteristics, but they are not actual governmental entities.

Question 39

A hedge fund with $225 million of initial capital charges a management fee of 1% and an incentive fee of 10%. The management fee is based on assets under management at year-end, and the incentive fee is calculated independently from the management fee. Assuming the fund earns a 15% return at year-end, total fees earned by the hedge fund during the year are closest to:

Correct answer: C

A. $5.63 million. - Incorrect

Incorrect Answer Feedback: Incorrect. $5.63 million results from calculating the management fee based on assets under management at the beginning of the period, rather than the end of the period: Year-end value = $225 million × 1.15 = $258.75 million Management fee = Beginning value × Management fee % = $225 million × 1% = $2.25 million Incentive fee = (Year-end value – Beginning value) × Incentive fee % = ($258.75 million – $225 million) × 10% = $3.375 million Total fees = Management fee + Incentive fee = $2.25 million + $3.375 million = $5.625 million = $5.63 million

B. $5.70 million. - Incorrect

Incorrect Answer Feedback: Incorrect. $5.70 million represents the total fees earned if the incentive fee was calculated net of the management fee as opposed to independently from the management fee: Year-end value = $225 million × 1.15 = $258.75 million Management fee = Year-end value × Management fee % = $258.75 million × 1% = $2.5875 million Incentive fee = (Year-end value – Beginning value – Management fee) × Incentive fee % = ($258.75 million – $225 million – $2.5875 million) × 10% = $3.1163 million Total fees = Management fee + Incentive fee = $2.5875 million + $3.1163 million = $5.7038 million = $5.70 million

C. $5.96 million. - Correct

Correct Answer Feedback: Correct. Total fees earned by the hedge fund are closest to $5.96 million: Year-end value = $225 million × 1.15 = $258.75 million Management fee = Year-end value × Management fee % = $258.75 million × 1% = $2.5875 million Incentive fee = (Year-end value – Beginning value) × Incentive fee % = ($258.75 million – $225 million) × 10% = $3.375 million Total fees = Management fee + Incentive fee = $2.5875 million + $3.375 million = $5.9625 million = $5.96 million

Question 40

Which of the following statements is most accurate?

Correct answer: B

A. The general partner theoretically has limited liability - Incorrect

Incorrect Answer Feedback: Incorrect because in the world of alternative investments, partnership structures are common. In limited partnerships, the fund manager is the general partner (GP) and investors are the limited partners (LPs). LPs, who are generally accredited investors (owing to legal restrictions on the fund), are expected to understand and be able to assume the risks associated with the investments, which are less regulated than offerings to the general public. The GP runs the business and theoretically bears unlimited liability for anything that goes wrong.

B. Co-investment rights allow the limited partner to make additional direct investments in portfolio companies - Correct

Correct Answer Feedback: Correct because LPs play passive roles and are not involved with the management of the fund (although co-investment rights allow for the LPs to make additional direct investments in the portfolio companies); the operations and decisions of the fund are controlled solely by the GP.

C. The side letter is a legal document that outlines the rules of the partnership between the general partners and all limited partners - Incorrect

Incorrect Answer Feedback: Incorrect because the partnership between the GP and LPs is governed by a limited partnership agreement (LPA), a legal document that outlines the rules of the partnership and establishes the framework that ultimately guides the fund's operations throughout its life. In addition to LPAs, side letters may also be negotiated. Side letters are side agreements created between the GP and a certain number of LPs that exist outside the LPA.

Question 41

Which of the following statements about covered bonds is most accurate?

Correct answer: B

A. Covered bonds expose investors to prepayment risk. - Incorrect

Incorrect Answer Feedback: Incorrect because in contrast to a static pool of mortgage loans that expose investors to prepayment risk as in the case of US mortgage-backed securities, cover pool sponsors must replace any prepaid or non-performing assets (i.e., assets that do not generate the promised cash flows) in the cover pool to ensure sufficient cash flows until the maturity of the covered bond.

B. Covered bonds usually consist of one bond class per cover pool. - Correct

Correct Answer Feedback: Correct because whereas ABS often use credit tranching to create bond classes with different borrower default exposures, covered bonds usually consist of one bond class per cover pool.

C. The cover pool is transferred to a bankruptcy-remote special legal entity. - Incorrect

Incorrect Answer Feedback: Incorrect because in the case of ABS, the financial institution that originates loans transfers securitized assets to a bankruptcy-remote special legal entity, but the pool of underlying assets in a covered bond (or "cover pool") remains on the financial institution's balance sheet, against which covered bondholders retain a top-priority claim.

Question 42

An analyst gathers the following information about a float-adjusted market-capitalization- weighted index composed of three stocks: Shares % of Shares in Market Stock Price Outstanding Float 1 8,500 100 $42 2 10,000 75 $69 3 4,000 100 $95 Stock 1's weighting in the index is closest to:

Correct answer: C

A. 20.4%. - Incorrect

Incorrect Answer Feedback: Incorrect because it is the price-weighted weight: Price weight = price / sum of all prices in index = 42 / (42 + 69 + 95) = 42 / 206 = ≈ 0.204 = 20.4%.

B. 25.0%. - Incorrect

Incorrect Answer Feedback: Incorrect because it fails to take into account the float-adjustment: Market-cap weight = market cap / sum of all market caps in index = 42 × 8,500 / (42 × 8,500 + 69 × 10,000 + 95 × 4000) = 357,000 / (357,000 + 690,000 + 380,000) = 357,000 / 1,427,000 ≈0.25 = 25.0%.

C. 28.5%. - Correct

Correct Answer Feedback: Correct because Stock 1's float-adjusted market-capitalization weighting is calculated: = Float-adjusted market capitalization / sum of all float-adjusted market caps in index = 42 × 8,500 × 100% / (42 × 8,500 × 100% + 69 × 10,000 × 75% + 95 × 4000 × 100%) = 357,000 / (357,000 + 517,500+ 380,000) = 357,000 / 1,254,500 ≈ 0.285 = 28.5%.

Question 43

A trader bought shares for $20 per share at an initial margin of 40%. If the maintenance margin is 20%, the price below which the first margin call occurs is:

Correct answer: B

A. $10. - Incorrect

Incorrect Answer Feedback: Incorrect because it calculates the margin call price to be: (1 – 0.4)/(1 + 0.2) × $20 = $10.

B. $15. - Correct

Correct Answer Feedback: Correct because initial margin = $20 × 40% = $8. When the final price of the stock is P, the change in trader's equity = $8 + (P – $20) = (P – $12). For a margin call to happen, total equity per share / total price per share = (P – $12)/P < 20 % (maintenance margin). Hence P < $15.

C. $16. - Incorrect

Incorrect Answer Feedback: Incorrect because it uses 20% for initial margin instead of 40%. Initial margin = $20 × 20% = $4, and it calculates the margin call price to be $20 – $4 = $16.

Question 44

An analyst gathers the following information about zero rates: Years to Zero Rate Maturity 1 1.50% 2 2.00% 3 2.25% The 2-year implied forward rate in one year's time is closest to:

Correct answer: B

A. 2.50%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the implied forward rate that starts in year 1 and ends in year 2 with a tenor of 1 (= 2 – 1) year, denoted with IFR , whereas what is 1,1 being asked is IFR . The general formula for the relationship between the two spot rates (z , 1,2 A z) and the implied forward rate (IFR) that starts in period A and ends in period B is given B A,B–A by (1 + z)A × (1 + IFR)B–A = (1 + z)B A A,B–A B Therefore, for the provided information: (1 + z)1 × (1 + IFR)2–1 = (1 + z)2, 1 1,2–1 2 (1 + 1.5%)1 × (1 + IFR)1 = (1 + 2%)2, 1,1 IFR ≠ IFR = (1 + 2%)2/(1 + 1.5%)1 – 1 = 0.025025 ≈ 2.50% 1,2 1,1

B. 2.63%. - Correct

Correct Answer Feedback: Correct because this is the implied forward rate that starts in year 1 and ends in year 3 with a tenor of 2 (= 3 – 1) years, denoted as IFR . The general formula for the 1,2 relationship between the two spot rates (z , z) and the implied forward rate (IFR) that A B A,B–A A B–A B starts in period A and ends in period

B is given by (1 + z) × (1 + IFR) = (1 + z) A A,B–A B Therefore, for the provided information: (1 + z)1 × (1 + IFR)3–1 = (1 + z)3, 1 1,3–1 3 (1 + 1.5%)1 × (1 + IFR)2 = (1 + 2.25%)3, 1,2 IFR = [(1 + 2.25%)3/(1 + 1.5%)1]1/2 – 1 = 0.026271 ≈ 2.63% 1,2

C. 2.75%. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the implied forward rate zero-coupon bond that starts in year 2 and ends in year 3 with a tenor of 1 (= 3 – 2) year, denoted with IFR , 2,1 whereas what is being asked is IFR . The general formula for the relationship between the two 1,2 spot rates (z , z) and the implied forward rate (IFR) that starts in period A and ends in A B A,B–A period B is given by (1 + z)A × (1 + IFR)B–A = (1 + z)B A A,B–A B Therefore, for the provided information: (1 + z)2 × (1 + IFR)3–2 = (1 + z)3, 2 2,3–2 3 2 1 3 (1 + 2%) × (1 + IFR) = (1 + 2.25%) , 2,1 3 2 IFR ≠ IFR = (1 + 2.25%) /(1 + 2%) – 1 = 0.027518 ≈ 2.75% 1,2 2,1

Question 45

If a portfolio consists of two risky assets and the portfolio's standard deviation is equal to the weighted average of the assets' standard deviations, the assets' returns are:

Correct answer: C

A. perfectly negatively correlated. - Incorrect

Incorrect Answer Feedback: Incorrect because when two assets in a portfolio are perfectly negatively correlated or ρ = –1, the portfolio's standard deviation is: 12 σ p = which is less than the weighted average of the securities' standard deviations.

Incorrect Answer Feedback: Incorrect because when two assets in a portfolio are perfectly negatively correlated or $\rho_{12} = -1$, the portfolio's standard deviation is:

$$\sigma_p = \sqrt{\omega^2{}_1\sigma^2{}_1 + \omega^2{}_2\sigma^2{}_2 - 2\omega_1\omega_2\sigma_1\sigma_2}$$

which is less than the weighted average of the securities' standard deviations.

Figure 11: Source equation for question 45, option A

B. uncorrelated. - Incorrect

Incorrect Answer Feedback: Incorrect because when two assets in a portfolio are uncorrelated or , ρ = 0 the portfolio's standard deviation is: σ which is less than 12 p = the weighted average of the securities' standard deviations.

Incorrect Answer Feedback: Incorrect because when two assets in a portfolio are uncorrelated or , $\rho_{12} = 0$ the portfolio's standard deviation is: $\sigma_p = \sqrt{\omega^2{}_1\sigma^2{}_1 + \omega^2{}_2\sigma^2{}_2}$ which is less than the weighted average of the securities' standard deviations.

Figure 12: Source equation for question 45, option B

C. perfectly positively correlated. - Correct

Correct Answer Feedback: Correct because portfolio risk is unaffected when the two assets are perfectly correlated or ρ = +1, the portfolio standard deviation is: 12 σ p = In other words, the portfolio's standard deviation is simply a weighted average of the standard deviations of the two assets and as such a portfolio's risk is unchanged with the addition of assets with the same risk parameters.

Correct Answer Feedback: Correct because portfolio risk is unaffected when the two assets are perfectly correlated or $\rho_{12} = +1$, the portfolio standard deviation is:

$$\sigma_p = \sqrt{\omega^2{}_1\sigma^2{}_1 + \omega^2{}_2\sigma^2{}_2 + 2\omega_1\omega_2\sigma_1\sigma_2}$$

In other words, the portfolio's standard deviation is simply a weighted average of the standard deviations of the two assets and as such a portfolio's risk is unchanged with the addition of assets with the same risk parameters.

Figure 13: Source equation for question 45, option C

Question 46

Which of the following is most likely associated with secondary capital markets?

Correct answer: B

A. Book building - Incorrect

Incorrect Answer Feedback: Incorrect. Book building is associated with primary markets.

B. Continuous trading - Correct

Correct Answer Feedback: Correct. A secondary market is where investors continue to trade the securities among themselves. Continuous trading is typically associated with secondary capital markets.

C. Lead underwriters - Incorrect

Incorrect Answer Feedback: Incorrect. Lead underwriter is associated with primary markets.

Question 47

Which of the following sections of the IPS most likely explains how to respond to various contingencies?

Correct answer: A

A. Procedures - Correct

Correct Answer Feedback: Correct because the procedures section explains the steps to take to keep the IPS current and the procedures to follow to respond to various contingencies.

B. Investment guidelines - Incorrect

Incorrect Answer Feedback: Incorrect because the investment guidelines section provides information about how policy should be executed and on specific types of assets excluded from investment, if any.

C. Evaluation and review - Incorrect

Incorrect Answer Feedback: Incorrect because this evaluation and review section provides guidance on obtaining feedback on investment results.

Question 48

If a test rejects the hypothesis that market prices reflect private information but does not reject the hypothesis that they reflect past market data and public information, then the form of market efficiency is best described as:

Correct answer: C

A. weak. - Incorrect

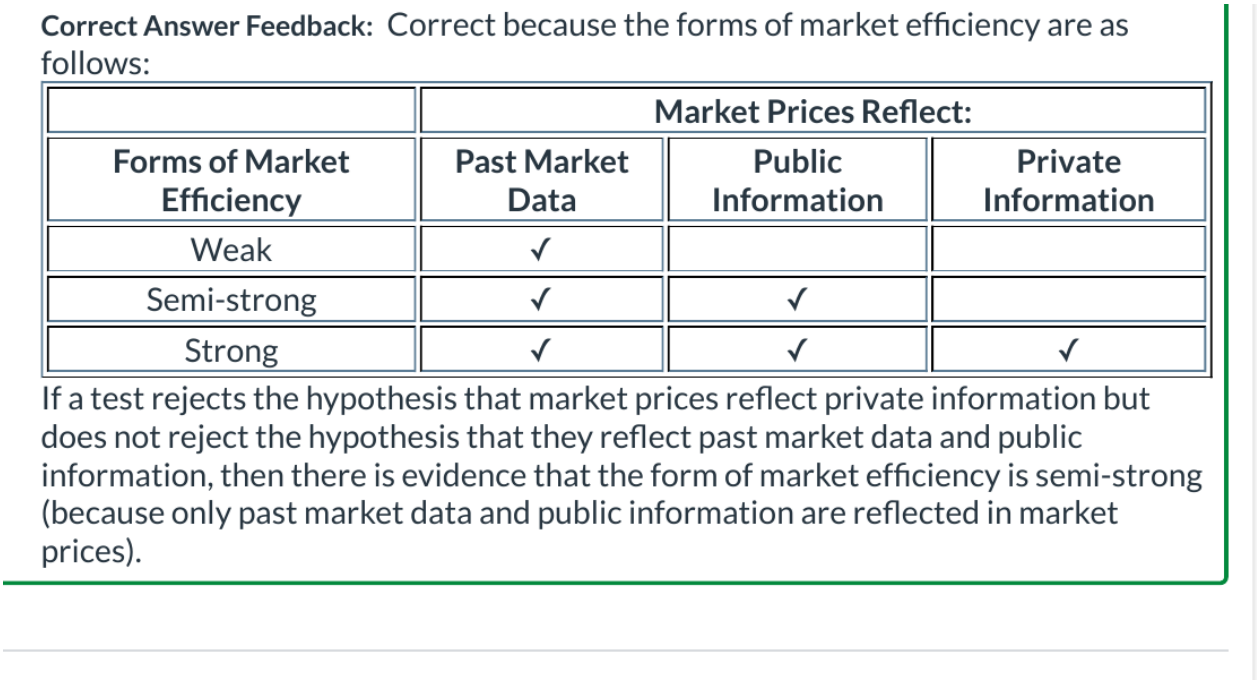
Incorrect Answer Feedback: Incorrect because markets are weak-form efficient when market prices reflect past market data only.

B. strong. - Incorrect

Incorrect Answer Feedback: Incorrect because markets are strong-form efficient when market prices reflect past market data, public information, and private information.

C. semi-strong. - Correct

Correct Answer Feedback: Correct because the forms of market efficiency are as follows: Market Prices Reflect: Forms of Market Past Market Public Private Efficiency Data Information Information Weak ✓ Semi-strong ✓ ✓ Strong ✓ ✓ ✓ If a test rejects the hypothesis that market prices reflect private information but does not reject the hypothesis that they reflect past market data and public information, then there is evidence that the form of market efficiency is semi-strong (because only past market data and public information are reflected in market prices).



Correct Answer Feedback: Correct because the forms of market efficiency are as follows:

	Market Prices Reflect:		
Forms of Market Efficiency	**Past Market Data**	**Public Information**	**Private Information**
Weak	✓		
Semi-strong	✓	✓	
Strong	✓	✓	✓

If a test rejects the hypothesis that market prices reflect private information but does not reject the hypothesis that they reflect past market data and public information, then there is evidence that the form of market efficiency is semi-strong (because only past market data and public information are reflected in market prices).

Figure 14: Source equation for question 48, option C

Question 49

In order to take advantage of short-term capital market expectations, a fund manager holds less of her portfolio in equities than the policy weights prescribe. This action is best described as:

Correct answer: C

A. security selection. - Incorrect

Incorrect Answer Feedback: Incorrect because security selection is an attempt to generate higher returns than the asset class benchmark by selecting securities with a higher expected return. The action to reduce the weight of equities does not involve the selection of specific securities. It is tactical asset allocation, not security selection, that chooses asset class weights that deviate from the policy.

B. portfolio rebalancing. - Incorrect

Incorrect Answer Feedback: Incorrect because rebalancing refers to the process of restoring the portfolio's original exposures to systematic risk factors. However, the action here is to choose portfolio weightings that deviate from the original exposure, which is tactical asset allocation, not rebalancing.

C. tactical asset allocation. - Correct

Correct Answer Feedback: Correct because tactical asset allocation is the decision to deliberately deviate (e.g., reduce the weight of equities) from the policy exposures to systematic risk factors (i.e., the policy weights of asset classes) with the intent to add value based on forecasts of the near-term returns of those asset classes.

Question 50

An investor collects the following information about a put option:

Stock price at initiation	$110
Strike price	$105
Option premium	$4.50

At expiration, if the price of the stock is $100, the investor's profit from selling the put is:

Correct answer: A

A. –$0.50. - Correct

Correct Answer Feedback: Correct because Π = –Max(0,X – S) + p (profit to the put T 0 seller), where S is the price of the underlying at expiration, X is the strike price and p is the T 0 option premium. Therefore, the correct calculation yields: –$0.5 = –Max(0,$105 – $100) + $4.5.

B. $4.50. - Incorrect

Incorrect Answer Feedback: Incorrect because the profit for the put seller is not equal to p , the 0 option premium earned, or $4.5, since the put option expires in the money. Instead, the correct formula is Π = –Max(0,X – S) + p (profit to the put seller), where S is the price of the underlying T 0 T at expiration, X is the strike price and p is the option premium. Therefore, the correct calculation 0 yields: –$0.5 = –Max(0,$105 – $100) + $4.5.

C. $9.50. - Incorrect

Incorrect Answer Feedback: Incorrect because the profit for the put seller (Π) is not equal to Max(0,X – S) + p , where S is the price of the underlying at expiration, X is the strike price and T 0 T p is the option premium. Therefore, the incorrect calculation yields: $9.5 = Max(0,$105 – $100) 0 + $4.5. Instead, the correct formula is Π = –Max(0,X – S) + p (profit to the put seller) which T 0 yields: –$0.5 = –Max(0,$105 – $100) + $4.5.

Question 51

A Eurodollar bond is:

Correct answer: A

A. denominated in USD. - Correct

Correct Answer Feedback: Correct because Eurodollar bonds are denominated in US dollars.

B. sold to US investors at issuance. - Incorrect

Incorrect Answer Feedback: Incorrect because Eurobonds denominated in US dollars cannot be sold to US investors at the time of issue because they are not registered with the US Securities and Exchange Commission (SEC).

C. underwritten by a syndicate of US-based banks. - Incorrect

Incorrect Answer Feedback: Incorrect because Eurobonds are issued outside the jurisdiction of any single country. They are underwritten by an international syndicate—that is, a group of financial institutions from different jurisdictions—and mostly sold to investors in Europe, the Middle East, and Asia.

Question 52

When markets are operating normally, the portfolio approach to investing most likely:

Correct answer: A

A. affects an investor's risk more than the investor's return. - Correct

Correct Answer Feedback: Correct because the portfolio approach provides investors with a way to reduce the risk associated with their wealth without necessarily decreasing their expected rate of return.

B. provides the same level of risk reduction as during times of severe market turmoil. - Incorrect

Incorrect Answer Feedback: Incorrect because although portfolio diversification generally does reduce risk, it does not necessarily provide the same level of risk reduction during times of severe market turmoil as it does when the economy and markets are operating "normally".

C. increases an investor's expected return as compared to owning one individual stock. - Incorrect

Incorrect Answer Feedback: Incorrect because the expected return of a portfolio should not be higher or lower as compared to holding one randomly selected stock.

Question 53

Which of the following components of an IPS most likely states a client's requirements for withdrawing funds from the portfolio?

Correct answer: B

A. Statement of Purpose - Incorrect

Incorrect Answer Feedback: Incorrect because the Statement of Purpose section states the purpose of the IPS, and not the specific requirements of the clients.

B. Investment Constraints - Correct

Correct Answer Feedback: Correct because the IPS should state what the likely requirements are to withdraw funds from the portfolio. When the client does have such a requirement, the manager should allocate part of the portfolio to cover the liability. This is a liquidity constraint that would be reflected in the investment constraints component of the IPS. Major types of constraints on portfolio selection are liquidity, time horizon, tax concerns, legal and regulatory factors, and unique circumstances. These constraints are typically discussed in a section of the IPS called, Investment Constraints. This section presents the factors that constrain the client in seeking to achieve the investment objectives.

C. Statement of Duties and Responsibilities - Incorrect

Incorrect Answer Feedback: Incorrect because the Statement of Duties and Responsibilities section details the duties and responsibilities of the client, the custodian of the client's assets, and the investment managers, and not the specific requirements of the clients.

Question 54

The FCFE model:

Correct answer: B

A. requires an estimate of future dividend payments. - Incorrect

Incorrect Answer Feedback: Incorrect because in practice, many analysts prefer to use a free- cash-flow-to-equity (FCFE) valuation model. These analysts assume that dividend-paying capacity should be reflected in the cash flow estimates rather than expected dividends. FCFE is a measure of dividend-paying capacity.

B. can be used to value non-dividend paying stocks. - Correct

Correct Answer Feedback: Correct because analysts may also use FCFE valuation models for a non-dividend-paying stock.

C. treats non-cash expense in the same manner as cash expense. - Incorrect

Incorrect Answer Feedback: Incorrect because the calculation of FCFE starts with the calculation of cash flow from operations (CFO). CFO is simply defined as net income plus non- cash expenses minus investment in working capital. Cash expenses are included in net income but non-cash expenses are added back in calculating CFO.

Question 55

A 5-year option-free bond is priced at a discount to par value. Assuming the discount rate and other key variables do not change, one year from now the value of the bond will most likely:

Correct answer: C

A. decrease. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond's value must increase over time to be equal to its par value.

B. be the same. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond's value must increase over time to be equal to its par value.

C. increase. - Correct

Correct Answer Feedback: Correct because the bond is priced below its par value but will be worth exactly par value at maturity. Over time, assuming a stable discount rate, the value of the bond must increase so that it is equal to par at maturity. That is, the price is "pulled to par."

Question 56

Which of the following is categorized as a social infrastructure asset?

Correct answer: B

A. Airport - Incorrect

Incorrect Answer Feedback: Incorrect because an airport is a transportation asset and transportation assets are classified as economic (not social) infrastructure assets.

B. Correctional facility - Correct

Correct Answer Feedback: Correct because social infrastructure assets are directed toward human activities and include such assets as educational, health care, social housing, and correctional facilities, with the focus on providing, operating, and maintaining the asset infrastructure.

C. Telecommunciation tower - Incorrect

Incorrect Answer Feedback: Incorrect because a telecommunication tower is an information and communications technology (ICT) asset and ICT assets are classified as economic (not social) infrastructure assets.

Question 57

All else being equal, commodity forward prices will increase as a result of a decrease in the:

Correct answer: C

A. risk-free rate. - Incorrect

Incorrect Answer Feedback: Incorrect because all else being equal, a decreased risk-free rate will decrease the forward price, not increase it. The difference in prices between the cash or spot markets and derivative markets is equal to the cost of carry, which is the opportunity cost of holding these assets, and mirrors the risk-free rate and the cost of storing, transporting, and insuring the commodity. The holder of the physical commodity should expect to be compensated via a higher forward price, F +(T), than the prevailing cash price. 0

B. storage costs. - Incorrect

Incorrect Answer Feedback: Incorrect because all else being equal, decreased storage costs will decrease the forward price, not increase it. The difference in prices between the cash or spot markets and derivative markets is equal to the cost of carry, which is the opportunity cost of holding these assets, and mirrors the risk-free rate and the cost of storing, transporting, and insuring the commodity. The holder of the physical commodity should expect to be compensated via a higher forward price, F +(T), than the prevailing cash price. 0

C. convenience yield. - Correct

Correct Answer Feedback: Correct because since convenience yield is a benefit and accrues to the owner, it reduces the forward price. Thus, a decrease in convenience yield will increase the forward price.

Question 58

An analyst gathers the following credit measures for two companies and their industry: Credit Ratio Company 1 Company 2 Industry Operating profit margin 13.7% 12.1% 11.8% Debt/EBITDA 1.21x 2.89x 3.00x EBITDA/interest expense 15.6x 6.9x 6.9x Using these ratios, credit quality is strongest for:

Correct answer: A

A. Company 1. - Correct

Correct Answer Feedback: Correct because Company 1 has the best ratios for both EBITDA/interest expense (highest) and for Debt/EBITDA (lowest). Therefore, Company 1 has the strongest credit quality overall. EBITDA/Interest expense: This ratio measures the degree to which operating profit covers periodic interest payments, with a higher coverage ratio representing less credit risk. In some cases, the numerator is changed to include depreciation and amortization (EBITDA) as well as rental expense (EBITDAR), while the denominator may be increased to include lease payments or reduced by interest earned on marketable securities and cash. Debt to EBITDA: Leverage ratios are often expressed with debt either in the numerator or denominator, so care must be taken to reflect whether a greater or lesser number reflects increased leverage. For example, a higher debt-to-EBITDA ratio represents higher leverage, and debt may be adjusted to reflect operating leases or off-balance-sheet commitments. Rating agencies often use it as a trigger for rating actions, and banks reference it in loan covenants. A higher ratio indicates more leverage and thus higher credit risk. Note that this ratio can be very volatile for companies with high cash flow variability, such as those in cyclical industries and with high operating leverage (fixed costs). Company 1 has the best ratios for both EBITDA/interest expense (highest) and for Debt/EBITDA (lowest). Therefore, Company 1 has the strongest credit quality overall.

B. Company 2. - Incorrect

Incorrect Answer Feedback: Incorrect because, compared to Company 1, Company 2 has worse ratios for EBITDA/interest expense and for Debt/EBITDA.

C. the industry. - Incorrect

Incorrect Answer Feedback: Incorrect because, compared to Company 1, the industry has worse ratios for EBITDA/interest expense and for Debt/EBITDA.

Question 59

The bond equivalent yield for a semiannual pay bond is most likely:

Correct answer: C

A. equal to the effective annual yield. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond equivalent yield for a semi- annual pay bond is equal to double the semiannual yield to maturity and is lower than the effective annual yield.

B. more than the effective annual yield. - Incorrect

Incorrect Answer Feedback: Incorrect because the bond equivalent yield for a semi- annual pay bond is equal to double the semiannual yield to maturity and is lower than the effective annual yield.

C. equal to double the yield per semiannual period. - Correct

Correct Answer Feedback: Correct because an annual rate having a periodicity of two is known as a semiannual bond basis yield, or semiannual bond equivalent yield. Therefore, a semiannual bond basis yield is the yield per semiannual period times 2.

Question 60

Which of the following is least likely a feature of a credit card receivable asset-backed security (ABS)?

Correct answer: B

A. A lockout period - Incorrect

Incorrect Answer Feedback: Incorrect because a credit card receivable ABS would typically have a lockout period during which the cash flow that is paid out to security holders is based only on finance charges collected and fees.

B. Amortizing collateral - Correct

Correct Answer Feedback: Correct because a credit card receivable ABS is an example of an ABS with a non-amortizing collateral.

C. An early amortization provision - Incorrect

Incorrect Answer Feedback: Incorrect because a credit card receivable ABS may require early amortization of the principal if certain events occur. Such an early amortization provision would safeguard the credit quality of the issue.

Question 61

Prior to expiration, an at-the-money option most likely trades:

Correct answer: C

A. below intrinsic value. - Incorrect

Incorrect Answer Feedback: Incorrect because the price of an at-the-money option prior to expiration is above the option’s intrinsic value, as described in the justification for the correct answer.

B. at intrinsic value. - Incorrect

Incorrect Answer Feedback: Incorrect because the price of an at-the-money option prior to expiration is above the option’s intrinsic value, as described in the justification for the correct answer.

C. above intrinsic value. - Correct

Correct Answer Feedback: Correct because the value of a European call at expiration is also sometimes referred to as the exercise value or intrinsic value. When the underlying is precisely at the exercise price, the option is said to be at-the-money. American options possess every characteristic of European options and one additional trait: they can be exercised at any time prior to expiration. Early exercise cannot be required, so the right to exercise early cannot have negative value. Thus, American options cannot sell for less than European options. Therefore, for a European or American call or a put option that is at-the-money, the relationship between value or trading

price of the option and time is positive, which means that the option has a value or price that exceeds the intrinsic value.

Question 62

The weighted average of the time to receipt of a bond's promised future payments is best measured by:

Correct answer: C

A. effective duration. - Incorrect

Incorrect Answer Feedback: Incorrect because the effective duration of a bond is the sensitivity of the bond's price to a change in a benchmark yield curve, not to the change in the bond's own yield-to-maturity. Effective duration is a curve duration statistic in that it measures interest rate risk in terms of a change in the benchmark yield curve (ΔCurve).

B. modified duration. - Incorrect

Incorrect Answer Feedback: Incorrect because modified duration provides an estimate of the percentage price change for a bond given a change in its yield-to-maturity; it provides a linear estimate of the percentage price change.

C. Macaulay duration. - Correct

Correct Answer Feedback: Correct because Macaulay duration is a weighted average of the time to receipt of the bond's promised payments, where the weights are the shares of the full price that correspond to each of the bond's promised future payments. Macaulay duration is measured in terms of time periods, not in terms of a percentage price change.

Question 63

Convenience yield is best described as the:

Correct answer: B

A. opportunity cost of the money invested. - Incorrect

Incorrect Answer Feedback: Incorrect because there is the opportunity cost of the money invested. If a person buys an asset, he forgoes interest on his money. The effect on this interest is reflected by compounding the price paid for the asset to a future value at the risk-free rate of interest. However, convenience yield is not the opportunity cost of the money invested. Instead, convenience yield represents a nonmonetary advantage of holding the asset.

B. nonmonetary advantage of holding the asset. - Correct

Correct Answer Feedback: Correct because convenience yield represents a nonmonetary advantage of holding the asset.

C. net of the costs and benefits of holding an asset. - Incorrect

Incorrect Answer Feedback: Incorrect because the net of the costs and benefits is often referred to by the term carry, or sometimes cost of carry. By contrast, convenience yield represents a nonmonetary advantage of holding the asset.

Question 64

A stock currently priced at $75 has a dividend yield of 3%, based on the most recent dividend. If dividends are expected to grow by 8% annually, the stock's required rate of return is closest to:

Correct answer: B

A. 11.00%. - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly computes the required rate of return using the current dividend instead of the dividend expected next period.

Incorrect Answer Feedback: Incorrect because it incorrectly computes the required rate of return using the current dividend instead of the dividend expected next period.

$$r_e = \frac{\$2.25}{\$75} + 0.08 = 0.11 = 11\%$$

Figure 15: Source equation for question 64, option A

B. 11.24%. - Correct

Correct Answer Feedback: Correct because: Where: P = the current market value of the equity 0 D = the current dividend 0 D = the dividends next period 1 r = the required rate of return of the market e g = the expected growth rate of dividends D = current dividend = share price × dividend yield = $75 × 0.03 = $2.25. 0 D = D × (1 + g) = $2.25 × 1.08 = $2.43. 1 0

Correct Answer Feedback: Correct because:

$$P_0 = \frac{D_0(1+g)}{r_e - g} = \frac{D_1}{r_e - g}; P_0 = \frac{D_1}{r_e - g}; r_e = \frac{D_1}{P_0} + g$$

Where:

P_0 = the current market value of the equity

D_0 = the current dividend

D_1 = the dividends next period

r_e = the required rate of return of the market

g = the expected growth rate of dividends

D_0 = current dividend = share price × dividend yield = $75 × 0.03 = $2.25.

$D_1 = D_0 × (1 + g) = \$2.25 × 1.08 = \$2.43.$

$$r_e = \frac{\$2.43}{\$75} + 0.08 = 0.1124 = 11.24\%$$

Figure 16: Source equation for question 64, option B

C. 12.32%. - Incorrect

Incorrect Answer Feedback: Incorrect because it incorrectly inputs the current dividend yield, 3.00%, as the current dividend $3.00.

Incorrect Answer Feedback: Incorrect because it incorrectly inputs the current dividend yield, 3.00%, as the current dividend $3.00.

$$D_1 = \$3.00 \times (1 + 0.08) = \$3.24$$

$$r_e = \frac{\$3.24}{\$75} + 0.08 = 0.1232 = 12.32\%$$

Figure 17: Source equation for question 64, option C

Question 65

Which of the following is most likely a means of executing a differentiation competitive strategy?

Correct answer: A

A. Trademarks - Correct

Correct Answer Feedback: Correct because protection using trademarks, copyright, patents is a means of executing a differentiation strategy.

B. Economies of scope - Incorrect

Incorrect Answer Feedback: Incorrect because economies of scope is a means of executing a cost leadership strategy.

C. Geographical proximity to its customers - Incorrect

Incorrect Answer Feedback: Incorrect because proximity to customers and strong understanding of their needs is a means of executing a focus strategy.

Question 66

A limit order book (in € per share) has the following order prices:

Bids	Asks
-	48.20
-	48.00
47.85	-
47.70	-

A sell order is behind the market at a price of:

Correct answer: C

A. €47.70. - Incorrect

Incorrect Answer Feedback: Incorrect because a sell order at 47.70 would not be described as behind the market. Instead it would be the best offer at the market. A buy order at 47.70 would be described as behind the market.

B. €48.00. - Incorrect

Incorrect Answer Feedback: Incorrect because the lowest ask in the market is the best offer. A sell order at 48.00 would be the best offer (ask) at the market, not behind the market.

C. €48.20. - Correct

Correct Answer Feedback: Correct because a buy order placed below the best bid is behind the market. Similarly, a sell order that is above the best offer (ask) is said to be behind the market. So any sell order higher than 48.00 is behind the market.

Question 67

A trader gathers the following information about shares purchased on margin:

Purchase price per share	$30.00
Sale price per share	$25.00
Dividend per share	$0.50
Leverage ratio	2.50

If the shares were sold after one year, the total return is closest to:

Correct answer: B

A. –41.67%. - Incorrect

Incorrect Answer Feedback: Incorrect because dividend received was omitted in the calculation leading to incorrect value for the remaining equity after the sale. Calculation steps (per share): The leverage ratio of 2.5 means that the trader financed 40% (1 / 2.5) of the purchase price, hence: Loan = (1 – 0.4) × $30.00 = $18.00 Initial investment = $30.00 × 0.4 = $12.00 Trading loss = – ($30.00 – $25.00) = –$5.00 Remaining equity = $12.00 – $5.00 = $7.00 Return on initial investment = ($7.00 – $12.00) / $12.00 ≈ – 41.67%.

B. –37.50%. - Correct

Correct Answer Feedback: Correct because to determine the return on this investment, the candidate needs to find the initial equity and then calculate the equity remaining after the sale. Calculation steps (per share): The leverage ratio of 2.50 means that the trader financed 40% (1 / 2.5) of the purchase price, hence: Loan = (1 – 0.4) × $30.00 = $18.00 Initial investment = $30.00 × 0.4 = $12.00 Trading loss = – ($30.00 – $25.00) = – $5.00 Dividend received = $0.50 Remaining equity = $12.00 – $5.00 + $0.5 = $7.50 Return on initial investment = ($7.50 – $12.00) / $12.00 = –37.50%.

C. –25.00%. - Incorrect

Incorrect Answer Feedback: Incorrect because the leverage ratio was misinterpreted to arrive at incorrect financing level and incorrect initial investment. Calculation steps (per share): The leverage ratio of 2.50 means that the trader financed 40% (1 / 2.5) of the purchase price, but instead the candidate assumes that trader financed (1 – 0.4) of the purchase price, hence: Loan = 0.4 × $30.00 = $12.00 Initial investment = $30.00 × (1 – 0.4) = $18.00 Trading loss = – ($30.00 – $25.00) = – $5.00 Dividend received = $0.50 Remaining equity = $18.00 – $5.00 + $0.50 = $13.50 Return on initial investment = ($13.50 – $18.00) / $18.00 ≈ – 25.00%.

Question 68

Pooled investment vehicles that frequently trade at a significant discount to net asset value are most likely:

Correct answer: A

A. closed-end funds. - Correct

Correct Answer Feedback: Correct because closed-end funds generally trade at a discount to their net asset values. The discount reflects the expenses of running the fund and sometimes investor concerns about the quality of the management.

B. open-ended funds. - Incorrect

Incorrect Answer Feedback: Incorrect because open-ended funds issue new shares and redeem existing shares on demand, usually on a daily basis. The price at which a fund redeems and sells the fund's shares is based on the net asset value of the fund's portfolio.

C. exchange-traded funds. - Incorrect

Incorrect Answer Feedback: Incorrect because the prices at which ETFs trade rarely differ much from net asset values. The market price and net asset values of ETFs tend to converge.

Question 69

Which of the following is a characteristic of a well-functioning financial system?

Correct answer: B

A. Bid–ask spreads are high - Incorrect

Incorrect Answer Feedback: Incorrect because well-functioning financial systems are characterized by liquid markets in which the costs of trading—commissions, bid–ask spreads, and order price impacts—are low (operationally efficient markets).

B. Prices reflect fundamental values - Correct

Correct Answer Feedback: Correct because well-functioning financial systems are characterized by prices that reflect fundamental values so that prices vary primarily in response to changes in fundamental values and not to demands for liquidity made by uninformed traders (informationally efficient markets).

C. Prices vary primarily in response to demands for liquidity made by traders - Incorrect

Incorrect Answer Feedback: Incorrect because well-functioning financial systems are characterized by prices that reflect fundamental values so that prices vary primarily in response to changes in fundamental values and not to demands for liquidity made by uninformed traders (informationally efficient markets). Therefore the prices vary primarily due to fundamentals and not liquidity demands.

Question 70

The acquisition of a public company with the intent to take it private using debt financing best describes:

Correct answer: C

A. angel investing. - Incorrect

Incorrect Answer Feedback: Incorrect because pre-seed capital, or angel investing, is capital provided at the idea stage. Funds may be used to develop a business plan and to assess market potential. The amount of financing here is typically small and sourced from individuals, often friends and family, rather than by VC funds.

B. venture capital investing. - Incorrect

Incorrect Answer Feedback: Incorrect because venture capital entails investing in or providing financing to private companies with high growth potential. Typically, these are start-ups or young companies, but venture capital can be injected at various stages, ranging from concept creation for a company or near the point of a company's IPO (initial public offering) launch or its acquisition.

C. leveraged buyout investing. - Correct

Correct Answer Feedback: Correct because leveraged buyouts, or highly leveraged transactions, arise when private equity firms establish buyout funds (or LBO funds) to acquire public companies or established private companies, with a significant percentage of the purchase price financed through debt. The target company's assets typically serve as collateral for the debt, and the target company's cash flows are expected to be sufficient to service the debt. The debt becomes part of the target company's capital structure after the buyout occurs. After the transaction, the target company becomes or remains a privately owned company.

Question 71

A bond has a modified duration of 5 and a convexity statistic of 75. If the bond's yield to maturity decreases 50 bps, the expected percentage price change is closest to:

Correct answer: B

A. 2.41%. - Incorrect

Incorrect Answer Feedback: Incorrect because it subtracts the second term instead of adding it: = 0.025 − 0.00094 = 0.02406 ~ 2.41%.

Incorrect Answer Feedback: Incorrect because it subtracts the second term instead of adding it: $\left(-5 \times -0.005\right) - \left[\frac{1}{2} \times 75 \times (-0.005)^2\right]$ = 0.025 − 0.00094 = 0.02406 ~ 2.41%.

Figure 18: Source equation for question 71, option A

B. 2.59%. - Correct

Correct Answer Feedback: Correct because: = 0.025 + 0.00094 = 0.02594 ~ 2.59%.

Correct Answer Feedback: Correct because: $\%\triangle PV^{Full} \approx \left(AnnModDur \times \triangle Yield\right) + \left[\frac{1}{2} \times AnnConvexity \times (\triangle Yield)^2\right] = \left(-5 \times -0.005\right) - \left[\frac{1}{2} \times 75 \times (-0.005)^2\right]$ = 0.025 + 0.00094 = 0.02594 ~ 2.59%.

Figure 19: Source equation for question 71, option B

C. 2.69%. - Incorrect

Incorrect Answer Feedback: Incorrect because it omits 0.5 from the calculation: = 0.025 + 0.001875 = 0.026875 ~ 2.69%.

Incorrect Answer Feedback: Incorrect because it omits 0.5 from the calculation: $\left(-5 \times -0.005\right) - \left[75 \times (-0.005)^2\right]$ = 0.025 + 0.001875 = 0.026875 ~ 2.69%.

Figure 20: Source equation for question 71, option C

Question 72

Which of the following institutional investors typically has the lowest tolerance for risk?

Correct answer: C

A. University endowment funds - Incorrect

Incorrect Answer Feedback: Incorrect because the risk tolerance of university endowment funds is typically high.

B. Defined benefit pension plans - Incorrect

Incorrect Answer Feedback: Incorrect because the risk tolerance of defined benefit pension plans is typically quite high.

C. Property and casualty insurance companies - Correct

Correct Answer Feedback: Correct because insurance companies' risk tolerance is typically quite low. Insurance companies receive premiums for the policies they write, and they need to invest these premiums in a manner that will allow them to pay claims. To pay claims to policyholders, regulatory guidelines maintain that an insurance company's general account is typically invested conservatively in a diverse allocation of fixed- income securities.

Question 73

A put option has a strike price of $20.50. The option premium is $1.00. If the price of the underlying at expiration is $21.50, the put option is:

Correct answer: C

A. in the money. - Incorrect

Incorrect Answer Feedback: Incorrect because a put option is called "in the money" if the underlying value is less than the exercise price.

B. at the money. - Incorrect

Incorrect Answer Feedback: Incorrect because a put option is called "at the money" if the underlying value is equal to the exercise price.

C. out of the money. - Correct

Correct Answer Feedback: Correct because a put option is called "out of the money" if the underlying value is greater than the exercise price. In this case, the underlying value of $21.50 is greater than the strike price of $20.50.

Question 74

Momentum, defined as relative past stock returns, is most likely a factor in:

Correct answer: B

A. fundamental factor models. - Incorrect

Incorrect Answer Feedback: Incorrect because fundamental factor models analyze and use relationships between security returns and the company's underlying fundamentals, for example, earnings, earnings growth, cash flow generation, investment in research, advertising, and number of patents.

B. the Carhart four-factor model. - Correct

Correct Answer Feedback: Correct because Mark Carhart (1997) extended the Fama and French model by adding another factor: momentum, defined as relative past stock returns. The best example of a practical model is the four-factor model proposed by Fama and French (1992) and Carhart (1997).

C. the Fama–French three-factor model. - Incorrect

Incorrect Answer Feedback: Incorrect because Fama and French (1992) proposed that three factors seem to explain asset returns better than just systematic risk. Those three factors are relative size, relative book-to-market value, and beta of the asset. Mark Carhart (1997) extended the Fama and French model by adding another factor: momentum, defined as relative past stock returns.

Question 75

Two risk managers are discussing how an organization's risk tolerance should be determined. The first manager says, "The risk tolerance must reflect the losses or shortfalls that will cause the organization to fail to meet critical objectives." The second manager responds, "The risk tolerance must reflect the external forces that bring uncertainty to the organization." Which of them is most likely correct?

Correct answer: A

A. Both risk managers - Correct

Correct Answer Feedback: Correct because the risk tolerance of an organization should reflect both an “inside” view and an “outside” view. The inside view asks what level of loss will leave the organization unable to meet critical objectives. The outside view asks what sources of uncertainty or risk the organization faces.

B. The first risk manager - Incorrect

Incorrect Answer Feedback: Incorrect because both an “inside” and “outside” view must be reflected.

C. The second risk manager - Incorrect

Incorrect Answer Feedback: Incorrect because both an “inside” and “outside” view must be reflected.

Question 76

All else being equal, a stock split results in:

Correct answer: B

A. a decrease in the number of shares and an increase in the share price. - Incorrect

Incorrect Answer Feedback: Incorrect because this describes a reverse stock split which involves a reduction in the number of shares outstanding with a corresponding increase in share price.

B. an increase in the number of shares and a decrease in the share price. - Correct

Correct Answer Feedback: Correct because a stock split involves an increase in the number of shares outstanding with a consequent decrease in share price.

C. an increase in the number of shares and an increase in the share price. - Incorrect

Incorrect Answer Feedback: Incorrect because a stock split involves an increase in the number of shares outstanding with a consequent decrease in share price.

Question 77

Mass redemptions may force the sale of underlying real estate by managers of:

Correct answer: A

A. open-end funds only. - Correct

Correct Answer Feedback: Correct because the REIT is not forced to sell the company’s underlying real estate like open-end funds experiencing mass redemptions.

B. publicly traded REITs only. - Incorrect

Incorrect Answer Feedback: Incorrect because the REIT is not forced to sell the company’s underlying real estate like open-end funds experiencing mass redemptions.

C. both open-end funds and publicly traded REITs. - Incorrect

Incorrect Answer Feedback: Incorrect because the REIT is not forced to sell the company’s underlying real estate like open-end funds experiencing mass redemptions.

Question 78

With respect to behavioral finance, an explanation for the overreaction anomaly is:

Correct answer: B

A. risk aversion. - Incorrect

Incorrect Answer Feedback: Incorrect because loss aversion, not risk aversion is an explanation for the overreaction anomaly. Some argue that behavioral theories of loss aversion can explain observed overreaction in markets.

B. loss aversion. - Correct

Correct Answer Feedback: Correct because some argue that behavioral theories of loss aversion can explain observed overreaction in markets. If loss aversion is more important than risk aversion, researchers should observe that investors overreact.

C. sudden changes in economic fundamentals. - Incorrect

Incorrect Answer Feedback: Incorrect because finding that equity returns are affected by changes in economic fundamentals is not evidence of market inefficiency and would not result in abnormal trading returns. These changes do not explain the overreaction anomaly.

Question 79

An issuer credit rating most likely applies to the issuer's:

Correct answer: C

A. secured debt. - Incorrect

Incorrect Answer Feedback: Incorrect because an issuer credit rating is meant to address an obligor's overall creditworthiness—its ability and willingness to make timely payments of interest and principal on its debt. The issuer credit rating usually applies to its senior unsecured debt, not to its secured debt.

B. subordinated debt. - Incorrect

Incorrect Answer Feedback: Incorrect because an issuer credit rating is meant to address an obligor's overall creditworthiness—its ability and willingness to make timely payments of interest and principal on its debt. The issuer credit rating usually applies to its senior unsecured debt, not to its subordinated debt.

C. senior unsecured debt. - Correct

Correct Answer Feedback: Correct because an issuer credit rating is meant to address an obligor's overall creditworthiness—its ability and willingness to make timely payments of interest and principal on its debt. The issuer credit rating usually applies to its senior unsecured debt.

Question 80

Which of the following best describes the process of changing the distribution of risk outcomes by using derivatives?

Correct answer: A

A. Risk shifting - Correct

Correct Answer Feedback: Correct because risk shifting refers to actions that change the distribution of risk outcomes. Risk transfer is often associated with insurance, whereas risk shifting generally involves derivatives as the risk modification vehicle. Risk shifting diverts some portion of the risk distribution to another market participant who either bears the risk or intermediates that risk by moving it to yet another party.

B. Risk transfer - Incorrect

Incorrect Answer Feedback: Incorrect because risk transfer is the process of passing on a risk to another party, often, but not always, in the form of an insurance policy. Whereas risk transfer refers to actions taken that pass the risk on to other parties, risk shifting refers to actions that change the distribution of risk outcomes. Risk transfer is often associated with insurance, whereas risk shifting generally involves derivatives as the risk modification vehicle.

C. Risk avoidance - Incorrect

Incorrect Answer Feedback: Incorrect because one method of managing risk is taking steps to avoid it altogether; however, avoiding risk may not be as simple as it appears. It is difficult to completely avoid risk, but more importantly, it is unclear that every risk should be completely avoided particularly if there are high costs associated with eliminating the risk. On the other hand, risk shifting refers to actions that change the distribution of risk outcomes.

Question 81

Asset-based valuation models are best used for companies that have:

Correct answer: C

A. a high proportion of intangibles. - Incorrect

Incorrect Answer Feedback: Incorrect because asset-based valuations work well for companies that do not have a high proportion of intangible or “off the books” assets.

B. significant property, plant and equipment. - Incorrect

Incorrect Answer Feedback: Incorrect because companies with assets that do not have easily determinable market (fair) values—such as those with significant property, plant, and equipment —are very difficult to analyze using asset valuation methods.

C. a high proportion of current assets and current liabilities. - Correct

Correct Answer Feedback: Correct because asset-based valuations work well for companies that do not have a high proportion of intangible or “off the books” assets and that do have a high proportion of current assets and current liabilities.

Question 82

According to put-call-forward parity, a long put and a short call is equal to:

Correct answer: A

A. a long bond and a short forward. - Correct

Correct Answer Feedback: Correct because the following is referred to as put–call forward parity: F (T)(1+r)–T + p = c + X(1+r)-T, where F (T)(1+r)–T is the present value of the forward price 0 0 0 0 discounted at the risk-free rate r, p is the price of a put option with an exercise price of X at t=T, 0 c is the price of a call option with an exercise price of X at t=T, and X(1 + r)-T is the value of a 0 risk-free bond with a face value X, with X equal to the forward price F (T) that is discounted at 0 the risk-free rate r. Rearranging the equation yields: p – c = X(1+r)-T – F (T)(1+r)–T. Therefore, a long put and a 0 0 0 short call is equal to a long bond and a short forward.

B. a short bond and a long forward. - Incorrect

Incorrect Answer Feedback: Incorrect because the following is referred to as put–call forward parity: F (T)(1 + r)–T + p = c + X(1 + r)–T, where F (T)(1 + r)–T is the present value of the forward price 0 0 0 0 discounted at the risk-free rate r, p is the price of a put option with an exercise price of X at t = T, 0 c is the price of a call option with an exercise price of X at t = T, and X(1 + r)–T is the value of a 0 risk-free bond with a face value X, with X equal to the forward price F (T) that is discounted at 0 the risk-free rate. Rearranging the equation yields: p – c = X(1 + r)–T – F (T)(1 + r)–T. Therefore, a long put and a 0 0 0 short call is equal to a long bond and a short forward. It is not equal to a short bond and a long forward according to put–call forward parity.

C. a short bond and a short forward. - Incorrect

Incorrect Answer Feedback: Incorrect because the following is referred to as put–call forward parity: F (T)(1 + r)–T + p = c + X(1 + r)–T, where F (T)(1 + r)–T is the present value of the forward price 0 0 0 0 discounted at the risk-free rate r, p is the price of a put option with an exercise price of X at t = T, 0 c is the price of a call option with an exercise price of X at t = T, and X(1 + r)–T is the value of a 0 risk-free bond with a face value X, with X equal to the forward price F (T) that is discounted at 0 the risk-free rate. Rearranging the equation yields: p – c = X(1 +

r)–T – F (T)(1 + r)–T. Therefore, a long put and a 0 0 0 short call is equal to a long bond and a short forward. It is not equal to a short bond and a short forward according to put–call forward parity.

Question 83

Which of the following statements is most likely correct regarding the spot and forward curves. The spot curve:

Correct answer: A

A. can be calculated from the forward curve, and the forward curve can be calculated from the spot curve. - Correct

Correct Answer Feedback: Correct because the forward and spot curves are interconnected to each other. The spot curve can be calculated from the forward curve, and the forward curve can be calculated from the spot curve. Either curve can be used to value fixed-rate bonds.

B. can be calculated from the forward curve, but the forward curve cannot be calculated from the spot curve. - Incorrect

Incorrect Answer Feedback: Incorrect because the spot curve can be calculated from the forward curve, and the forward curve can be calculated from the spot curve.

C. cannot be calculated from the forward curve, but the forward curve can be calculated from the spot curve. - Incorrect

Incorrect Answer Feedback: Incorrect because the spot curve can be calculated from the forward curve, and the forward curve can be calculated from the spot curve.

Question 84

Companies that are candidates for a leveraged buyout most likely:

Correct answer: B

A. require start-up capital. - Incorrect

Incorrect Answer Feedback: Incorrect because a company requiring start-up capital would likely seek venture capital. Venture capital investments provide ‘seed’ or start-up capital whereas, a leveraged buyout (LBO) occurs when a group of investors uses a large amount of debt to purchase all of the outstanding common shares of a publicly traded company.

B. generate a high level of cash flow. - Correct

Correct Answer Feedback: Correct because a leveraged buyout (LBO) occurs when a group of investors (such as the company’s management or a private equity partnership) uses a large amount of debt to purchase all of the outstanding common shares of a publicly traded company. Companies that are candidates for these types of transactions generally have large amounts of undervalued assets (which can be sold to reduce debt) and generate high levels of cash flows (which are used to make interest and principal payments on the debt).

C. are experiencing a rapid deterioration in operations. - Incorrect

Incorrect Answer Feedback: Incorrect because a company experiencing rapid deterioration in operations is unlikely to be able to service the debt involved with a leveraged buyout. A company in such a situation may issue a private investment in public equity, or PIPE. This type of investment is generally sought by a public company that is in need of additional capital quickly. For example, a company may require a large investment of new equity funds in a short period of time because it has significant expansion opportunities, is facing high levels of indebtedness, or is experiencing a rapid deterioration in its operations.

Question 85

An analyst gathers the following information about a hedge fund:

Beginning-of-year assets under management (AUM)	$350 million
Management fee, based on year-end AUM before fees	2%
Incentive fee	20%
Hard hurdle rate	5%
Annual return before fees	10%

If the incentive fee is based on returns net of management fees, the incentive fee (in $ millions) is:

Correct answer: A

A. 1.96. - Correct

Correct Answer Feedback: Correct because the hurdle rate is a minimum rate of return, typically 8%, that the GP must exceed in order to earn the performance fee. GPs typically receive 20% of the total profit of the private equity fund net of any hard hurdle rate, in which case the GP earns fees on annual returns in excess of the hurdle rate, or net of the soft hurdle rate, in which case the fee is calculated on the entire annual gross return as long as the set hurdle is exceeded. Management Fee = $350 × 1.10 × 0.02 = $7.7 Hurdle Rate (in $) = $350 × 0.05 = $17.5 Profit on Hedge Fund = $350 × 0.10 = $35 Incentive Fee = ($35 – $7.7 – $17.5) × 0.20 = $1.96.

B. 5.46. - Incorrect

Incorrect Answer Feedback: Incorrect because the hurdle rate is not accounted for in the incentive fee calculation. Profit on Hedge Fund = $350 × 0.10 = $35 Management Fee = $350 × 1.10 × 0.02 = $7.7 Incentive Fee = ($35 – $7.7) × 0.20 = $5.46.

C. 7.00. - Incorrect

Incorrect Answer Feedback: Incorrect because neither the management fee nor the hurdle rate are accounted for in the incentive fee calculation. Profit on Hedge Fund = $350 × 0.10 = $35 Incentive Fee = $35 × 0.20 = $7.00.

Question 86

The risk-free rate is 5%, and the market risk premium is 8%. If the beta of TRL Corp. is 1.5, based on the capital asset pricing model (CAPM), the expected return of TRL's stock is closest to:

Correct answer: C

A. 9.5%. - Incorrect

Incorrect Answer Feedback: Incorrect because in this case the formula misuses the MRP so the expected return is computed as 0.05 + (0.08 – 0.05)(1.5) = 9.5%.

B. 15.5%. - Incorrect

Incorrect Answer Feedback: Incorrect because the formula swaps the RFR and the MRP so expected return is computed as 0.08 + (0.05)(1.5) = 15.5%.

C. 17.0%. - Correct

Correct Answer Feedback: Correct because using the CAPM relationship of E(R) = R + [E(R) – i f m R]β, we can estimate the expected return as: E(R) = 0.05 + (0.08)(1.5) = 17.0%. f i i

Question 87

In capital market theory, the optimal portfolio for an investor who borrows at the risk-free rate to invest in the market portfolio is situated:

Correct answer: B

A. above the capital allocation line. - Incorrect

Incorrect Answer Feedback: Incorrect because the optimal investor portfolio is always situated on the capital allocation line. The optimal investor portfolio is selected on the capital allocation line by overlaying the indifference curves that incorporate investor preferences.

B. to the right of the market portfolio on the capital allocation line. - Correct

Correct Answer Feedback: Correct because the location of an optimal investor portfolio depends on the investor's risk preferences. Some less risk-averse investors (i.e., with a high risk tolerance) may wish to accept even more risk because of the chance of higher return. Such an investor may borrow money to invest more in the risky portfolio. The optimal investor portfolio for such an investor will lie to the right of the optimal risky portfolio on the capital allocation line.

C. at the tangent of the capital allocation line and the Markowitz efficient frontier. - Incorrect

Incorrect Answer Feedback: Incorrect because this is the optimal risky portfolio, not the optimal investor portfolio. With the addition of the risk-free asset, we are able to narrow our selection of risky portfolios to a single optimal risky portfolio, P, which is at the tangent of (the dominant capital allocation line) CAL(P) and the efficient frontier of risky assets. Also, the two portfolios cannot be identical because the investor's risk preference determines a levered portfolio, which is situated to the right of the optimal risky portfolio on the capital allocation line.

Question 88

All else being equal, as investors' minimum required rate of return on equity increases, a company's cost of equity most likely:

Correct answer: B

A. decreases and its stock price increases. - Incorrect

Incorrect Answer Feedback: Incorrect because a company's cost of equity is often used as a proxy for the investors' minimum required rate of return. As such, the cost of equity would not decrease. Also, the stock price would not increase.

B. increases and its stock price decreases. - Correct

Correct Answer Feedback: Correct because a company's cost of equity is often used as a proxy for the investors' minimum required rate of return. As such, an increase in the minimum rate of return required by investors will cause the company's cost of equity to increase. And, regarding the stock price, if investors require a higher rate of return on equity than the company's cost of equity, they would sell their shares resulting in a decline in the company's share price.

C. increases and its stock price increases. - Incorrect

Incorrect Answer Feedback: Incorrect because the stock price would not increase, it would decrease: if investors require a higher rate of return on equity than the company's cost of equity, they would sell their shares resulting in a decline in the company's share price.

Question 89

With respect to fintech applications, the instantaneous transfer of collateral in the event of default is an example of:

Correct answer: B

A. tokenization. - Incorrect

Incorrect Answer Feedback: Incorrect because tokenization refers to the process of representing ownership rights to physical assets on a blockchain or distributed ledger.

B. smart contracts. - Correct

Correct Answer Feedback: Correct because DLT has the potential to accommodate ‘smart contracts,’ which are computer programs that self-execute on the basis of pre-specified terms and conditions agreed to by the parties to a contract. Examples of smart contract use are the automatic execution of contingent claims for derivatives and the instantaneous transfer of collateral in the event of default.

C. high-frequency trading. - Incorrect

Incorrect Answer Feedback: Incorrect because high-frequency trading (HFT) is a form of algorithmic trading that makes use of vast quantities of granular financial data (tick data, for example) to automatically place trades when certain conditions are met.

Question 90

Which of the following best describes the intrinsic value of a security? The intrinsic value of a security is:

Correct answer: C

A. the price at which the security can be bought or sold. - Incorrect

Incorrect Answer Feedback: Incorrect because market value is the price at which an asset can currently be bought or sold. Intrinsic value (sometimes called fundamental value) is, broadly speaking, the value that would be placed on it by investors if they had a complete understanding of the asset’s investment characteristics.

B. represented by the intersection of supply and demand for the security. - Incorrect

Incorrect Answer Feedback: Incorrect because the market value (not the intrinsic value) of an asset represents the intersection of supply and demand—the point that is low enough to induce at least one investor to buy while being high enough to induce at least one investor to sell.

C. the value placed by investors based on a complete understanding of the security’s investment characteristics. - Correct

Correct Answer Feedback: Correct because intrinsic value (sometimes called fundamental value) is, broadly speaking, the value that would be placed on an asset by investors if they had a complete understanding of the asset’s investment characteristics.